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ZTXPlainaAI an explainable deep learning framework for encrypted traffic anomaly detection in Zero Trust Networks

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Abstract

The rapid adoption of encrypted communication protocols has raised privacy levels but has also dealt a significant blow to traditional intrusion detection systems that rely on payload inspection for anomaly detection. This poses a monumental hurdle for Zero Trust Networks, as it requires persistent verification and a wide range of granular intrusion detection. Existing classical or deep learning-based solutions achieve fair performance. Nonetheless, they are severely constrained in their robustness to varying encryption protocols, interpretability for security experts, and susceptibility to adversarial attacks on changing test samples. These shortcomings highlight the need for a system that balances detection performance with transparency and adaptability. To address this challenge, this paper describes an explainable AI-based anomaly-detection framework, ZTXPlainaAI, for encrypted payloads in the context of Zero Trust Networks. Specifically, the framework uses EncXplainNet, a mixed deep learning model featuring CNNs to extract local features, GRUs to capture temporal ordering, and an attention mechanism for human-interpretable decision-making. Additionally, SHAP-based feature attribution enhances transparency and interpretability, providing post hoc explanations to analysts. An adaptive reinforcement and feedback loop that further enables the model to adapt to changing traffic conditions over time. Due to the meticulously curated methods used in our evaluation, EncXplainNet achieves fully explained underpinnings for its decision processes, outperforming state-of-the-art models in our extensive experiments on the CIC-IDS2019 encrypted traffic subset. The accuracy, F1-score and AUC we achieved are 0.96, 0.96, and 0.98, respectively. Finally, we demonstrate the robustness of our method through extensive ablation studies, showing its stability across unseen protocols, data-record noise, and adversarial attacks. As such, ZTXPlainaAI provides an explainable anomaly detection solution that meets ZTX Security's operational constraints by balancing accuracy, robustness, and explainability.

Keywords Encrypted traffic, Zero Trust Networks, Explainable AI, Anomaly detection, Deep learning



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1 Introduction

Although the interests of vital communication and its dependant services necessitated an accelerated set of features for privacy and confidentiality in the fabric of modern networks, with encrypted communication protocols leaving state-of-the-art in their dust, the very design parameters of the interest, the transmission of information and therefore observability information, which the dependent infrastructure on such technology has had the unintended side effect of severely degrading traditional insight and anomaly detection mechanisms used for security. Now that traffic is increasingly encapsulated, traditional payload-inspection-based IDSs are becoming less effective, leaving many blind spots in network defence. With the trend toward adopting Zero Trust Network (ZTN) architectures, which do not automatically trust any user or device (inside or outside the organisation) and always verify, there is increasing demand for fast anomaly-detection algorithms for encrypted traffic that can circumvent the limitations of decryption overhead. While initial traffic classification and anomaly detection have utilised conventional machine learning techniques, practical use has been constrained by the inflexibility of such methods in the face of evolving threats and their opacity [1, 2].

To overcome these challenges, recent work has begun to explore explainable artificial intelligence (XAI) methods that balance detection accuracy and interpretability for security analysts. Works such as Zebin et al. [12] and Singh et al. [49]. XAI methods for anomaly detection based on encrypted service calls are up-and-coming [64]. Islam et al. [4] focus solely on trust and provide examples of availability, integrity, and confidentiality; their work reminds us of the broader context of trust that must be built in secure systems, including interpretability. However, most methods developed to date either focus on protocol, e.g., DNS-over-HTTPS, or on static ML models that do not adapt over time. Furthermore, hardening against adversarial attacks or new-to-organisation encryption protocols — essential elements in any meaningful Zero Trust context — has received scant attention. The framework is proposed to fill such a gap by providing all three dimensions, i.e., high detection accuracy, interpretability, and robustness, for encrypted traffic anomaly detection.

To mitigate this gap, we propose a completely novel explainable anomaly detection framework, called ZTXPlainAI, based on Zero Trust Networks. At its heart is an explainable hybrid deep learning model: Encode: EncXplainNet, that utilises a CNNs–GRUs architecture for modelling local features and temporal dependencies, and an attention mechanism for interpretable decision making. Finally, SHAP-based feature attribution provides post hoc explanations with both intrinsic and global interpretations. Together, these design choices enable the model to achieve state-of-the-art performance in anomaly detection while also providing explainability that helps security analysts easily apply Zero Trust principles.

Task: To devise an explainable AI model which can classify if the traffic is encrypted or not, detect anomalies in it on the encrypted traffic based on industrial-oriented, non-whimsical constraints (i.e. without inspecting payload) and on real-world observations that IoT-MQTT communication generates encrypted traffic. The framework targets the main objectives: (i) high classification accuracy and robustness against a variety of encrypted protocols, (ii) human-understandable and interpretable rationale on the prediction of an anomaly dedicated to human analysts, and (iii) adaptability to changing network conditions using reinforcement and feedback loop mechanisms.

There are multiple novelties of this work. The hybrid architecture, constructed by incorporating CNNs, GRUs, and attention mechanisms in EncXplainNet, uniquely suits encrypted traffic flows by learning to set spatial and temporal contexts. Second, our explainability pipeline enables interpretability beyond feature attribution alone by introducing dual-layer explainability, providing real-time actionable transparency (using SHAP and attention). Note that we also highlight the importance of robustness by stress-testing under noise injection, adversarial perturbations in unseen protocols, input noise, and many other unseen perturbations, demonstrating significant robustness not reported in previous work. Finally, the analyst-guided feedback loop allows for adaptive refinement of automated detections and builds trust through human-in-the-loop analysis in Zero Trust environments.

A summary of this research contribution is as follows. It first presents ZTXPlainaAI, an end-to-end explainable framework for detecting anomalies in encrypted traffic. Secondly, it presents EncXplainNet: a high-accuracy yet explainable hybrid deep learning model that integrates CNNs, GRUs, attention, and SHAP explanations within a single framework, achieving high accuracy while maintaining a good level of interpretability. Third, it offers a comprehensive comparison with classical machine learning and deep learning baselines and demonstrates superior performance in accuracy (0.96), F1-score (0.96), and AUC (0.98). Fourth, it addresses limitations such as ablation studies, robustness testing, and comparisons with state-of-the-art methods to demonstrate reliability and generalizability. Fifth, the approach is designed to be practical and focused on Zero Trust, ensuring it can be used in real-world settings where such concepts are familiar.

The rest of this paper is organised as follows: In Sect. 2, we position our work in the context of related work, reviewing previous approaches to the analysis of encrypted traffic, anomaly detection, and explainable AI. Preliminaries are provided in Sect. 3, including the problem formulation (Sect. 3.1), background work on deep learning for encrypted traffic identification (Sect. 3.2), foundations of XAI (Sect. 3.3), Zero Trust principles (Sect. 3.4), and notations (Sect. 3.5). Section 4: Proposes the methodology contributing to the dataset, system overview, EncXplainNet model architecture, integration of explainability in attending embedding, reinforcement loop, training setup, evaluation strategy and the threat model. Experimental results are presented in Sect. 5, including dataset preparation, baselines, evaluation metrics, performance comparison, ablation studies, explainability outputs, robustness analysis, and benchmarking against state-of-the-art methods. Results are summarised in Sect. 6, which emphasises the principal outcomes and trade-offs between detection power, interpretability, and execution cost. Finally, Sect. 7 concludes the paper by outlining the work's contributions and future directions, including validation on a larger dataset, a usability study targeted at analysts, lightweight deployment optimisation, and defence against adversarial approaches.

Moving beyond the state of the art, this research contributes to the performance, interpretability, and robustness of encrypted traffic anomaly detection in a Zero Trust paradigm in a modular fashion, opening avenues for practical, explainable cybersecurity solutions.

2 Related work

The growing dependence on encryption and the integration of Zero Trust Network (ZTN) architectures have led some researchers to develop not just accurate but also interpretable and flexible anomaly detection frameworks. The classic approach of inspecting payloads is infeasible in encrypted environments, and this has driven the quest for artificial intelligence (AI) and explainable artificial intelligence (XAI) methods. In this section, we examine previous work across four interrelated topics: anomaly detection in encrypted traffic, the role of deep learning in Zero Trust, XAI for intrusion detection, and identify research gaps.

2.1 Encrypted traffic anomaly detection

By now, enabling end-to-end encryption has made it challenging to detect anomalies in the increasing volume of encrypted traffic. In this context, a hybrid CNN–GRU (i.e., convolutional neural networks with gated recurrent units) method was introduced by Bakhshi and Ghita [17] for time-series traffic classification. Similarly, Ibraheem et al. For example [18], discussed methods for feature selection to detect anomalies in HTTPS and noted that classifiers achieve better performance with appropriate feature engineering. Long and Zhang [19] developed a parallel, autoencoder-based deep learning framework for encrypted traffic detection, using L1 regularisation to improve anomaly detection accuracy. The absence of operational benchmarks and the scarcity of datasets are two significant limitations of AI-based detection of encrypted traffic, as shown in a systematic literature review by Ji et al. [20].

Other research developed deep learning methods specific to encrypted flows. For PF Sense log analysis, CNNs and LSTMs were used Fotiadou et al. [21], and feature-mining methods for encrypted malicious traffic were proposed by Wang and Thing [22]. Ikram et al. [23] and Yoshimura et al. [24] focused on ensemble learning and stacked feature extraction to improve robustness for high-dimensional data. Papadogiannaki and Ioannidis [25] provided a comprehensive survey of encrypted traffic analysis and defences, examining the trade-off between traffic privacy and security.

Other work [26, 27, 28, 29, 30] on innovations, such as adversarially regularised autoencoders, machine learning in the cloud, and online adaptive deep learning pipelines for dynamic anomaly detection, also directly relates to latent representation learning. Some of the new models emphasised flexibility and robustness Gonaygunta et al. [31]. Flexible DL anomaly detection [31], scalable frameworks [32], and systematic reviews relating detection in the encrypted space with Zero Trust and emerging networks [33, 34, 35, 36].

2.2 Deep learning in Zero Trust Networks

Zero Trust means that verification is continuous and access is contextual. Haque et al. Using deep learning and XAI, ZTA was applied to UAV security [1]. This works by using explainable models that can detect malware in encrypted traffic [2]. The work of Ajish [6, 8, 9] provides in-depth reviews of the role of AI within Zero Trust, demonstrating how adaptive anomaly detection, encryption optimisation, and identity-based access controls gain fortification through integration with AI. Comparative reviews of ZTA approaches have been presented in [13, 40], and ZTA's impact on cloud security has been analysed in [16].

Four of the studies explored very advanced employment contexts. Khan [34], Barach [35]. The work of [36] discusses methods for software-defined networking (SDN), whereas Tiwari et al. [37]. Adaptive Zero Trust: AI-driven Zero Trust ID system [38], Paul et al. [39]. Synergistic AI A-enabled Zero Trust user-device profiling and access control frameworks. Other researchers, such as [44], examined Zero Trust in 6G, and [45] elaborated on its implications for national security.

In this regard, some recent research papers have focused on utilising deep learning and explainable models for network security and are presented in the section below. Such approaches include Ozdem [8] who proposed modelling evolving network interactions through temporal graph networks and explainable AI to develop a real-time anomaly detection framework. Similarly, Qazi et al. [54] present a deep learning-based intrusion detection method within the Zero-Touch Network Security architectural framework to enable automated, adaptive switching of malicious traffic.

Other contributions have also focused on the orchestration and automation of Zero Trust Architecture (ZTA) policies—Hale et al. [42] studied Zero Trust with ML components in complex systems. Orchestration challenges were discussed by Cao et al. [41], Moreover, enterprise adoption strategies were outlined by Sunkara [47], highlighting practical yet useful barriers to scaling solutions. Javeed et al. present a federated learning-based zero-trust architecture for IoT in [48], with a special focus on utilising distributed intrusion detection. Collectively, these studies demonstrate that deep learning, as part of a Zero Trust architecture, enables adaptive, context-aware anomaly detection in rapidly evolving protected environments that are moving toward full encryption.

2.3 Explainable AI for intrusion detection

As security gains traction with AI, interpretability becomes even more critical. Nas-cita et al. [3, 57] analysed XAI for traffic classification, focusing on SHAP and LIME as popular methods. Islam et al. [4, 54] proposed IDS architectures combined with XAI, and Nkoro et al. [5, 64] and Nwakanma et al. [7] utilised the explainability aspect in IoT and connected vehicles. Zebin et al. [12] proposed an explainable IDS for DNS-over-HTTPS traffic that achieved a high classification rate and provided transparent feature attribution.

Mahbooba et al. [14] and Favour et al. [15] studied interpretable trust management. At the same time, Kaur and Gupta [10], Senevirathna et al. [11], and Hale et al. [42] explained the need for explainability in next-generation 5G/6G and Zero Trust systems. Singh et al. [49], Mohale and Obagbuwa [50], and Neupane et al. [51] conducted reviews of explainable IDS techniques, and Alabdulatif [52] and Hassan et al. [53] introduced ensemble and LIME-based IDS for VANETs.

In addition, recent work highlights the importance of interpretable learning across a wide range of fields. Almuqren et al. [55], Shtayat et al. [56], and Chen et al. [58] presented feature selection for IoT IDS. Alonge and Israel [59], Gorza Lczany and Rudzinski [60], and Bayrak [61] are concerned with an interpretable IDS for VANET, IoT, or UAV settings. Izuazu et al. [62], Memon et al. [63], and Hoang et al. [65] extended eX to VR, IoMT, and industrial control systems. Nguyen et al. [66] and Mutalib et al. [67] advocated for both robust and user-friendly explainability, supporting the necessity of transparency in AI security.

Furthermore, recent studies emphasise the role of explainable AI in zero trust and advanced network security. Chen et al. [43] and García-Teodoro et al. [46] highlighted secure and adaptive access control mechanisms. Keshk et al. [68], Li et al. [69], and Aziz et al. [70] explored explainable intrusion and anomaly detection across IoT and vehicular networks. Gummadi et al. [71], Pande and Khamparia [72], and Rahman et al. [73] proposed XAI-driven detection frameworks, while Charmet et al. [74] and Hussein and El-dosuky [75] demonstrated their application in cybersecurity systems. Nazat et al. [76], Yan et al. [77], and Ali and Kostakos [78] further advanced explainable models, and Capuano et al. [79] and Alketbi and Mehmood [80] provided comprehensive surveys on XAI for secure and transparent AI systems.

2.4 Research gaps

Despite great achievements, significant issues remain unresolved. Compounding the two weaknesses, multiple studies attempt to increase detection's accuracy without considering its robustness toward emerging encrypted protocols and adversarial perturbations [18, 22, 28]. In addition to differences, SHAP and LIME, though popular for model interpretation, are open questions in Zero Trust deployments with high speed [3, 50, 57]. Furthermore, while much of the literature on Zero Trust has been reviewed [13, 34, 38, 42, 47], most of these works remain at a conceptual level, lacking concrete deployment plans. Finally, little work connects explainable anomaly detection with adaptive feedback loops for encrypted traffic, which encourages the creation of ZTXPlainaAI.

To set the context for this work, we summarise representative contributions in encrypted traffic anomaly detection, Zero Trust architectures, and explainable IDS solutions in Table 1. The studies included in this review exemplify progress in enabling research, such as hybrid approaches combining deep models [17, 19], CNN–LSTM-based approaches [21], and explainable methods based on SHAP and LIME [49, 50]. Nevertheless, based on the research gap column, existing solutions [20, 25, 40] tend not to scale to real-time traffic or encrypted flows, do not generalise well to evolving protocols, or have mainly remained conceptual in Zero Trust deployments. These limitations directly inspire the design of ZTXPlainaAI, which incorporates hybrid feature extraction, temporal learning, and explainability — adaptive refinement to address the challenges of resilience, interpretability, and deployability in the context of Zero Trust Networks.

3 Preliminaries

This section briefly describes the key concepts that provide the necessary background for the proposed framework. It contains the problem formulation, deep learning approaches for analysing encrypted traffic, and the principles of explainable AI. It also elaborates on fundamental concepts of the Zero Trust Network and the notations adopted in this research to ensure the methodology and experimental analysis are clear in the following sections.

3.1 Problem formulation

In compliant enterprise networks with Zero Trust architecture, everything is denied by default: all users, all devices, and all network flows are never trusted. Since encrypted traffic renders passive deep packet inspection ineffective, this paradigm will require

Table 1 Comparative summary of selected related work on encrypted traffic anomaly detection and Zero Trust Networks

References	Methodology/model	Dataset	Key findings	Research gap
Bakhshi & Ghita (2021) [17]	Hybrid CNN–GRU for anomaly detection in encrypted traffic	NSL-KDD, UNSW-NB15, CIC-IDS-2017	Hybrid CNN–GRU outperforms single models, balancing latency and accuracy	Limited resilience to unseen encrypted protocols
Long & Zhang (2023) [19]	Parallel autoencoder-based framework with L1 feature selection	Custom encrypted traffic dataset	Efficient parallel feature extraction improves robustness and accuracy (99.9%)	The scalability of real-time feature extraction is not addressed.
Ji et al. (2024) [20]	Systematic literature review of AI-based encrypted anomaly detection	30 selected encrypted traffic studies	AI models are effective, but datasets and benchmarks are inconsistent	Lack of standard datasets and operational benchmarks
Fotiadou et al. (2021) [21]	CNN and LSTM for pfSense log-based anomaly detection	pfSense Suricata logs	CNN+LSTM classifiers show high accuracy in real-world logs	Model generalisation across different environments is untested
Papadogiannaki & Ioannidis (2021) [25]	Survey on encrypted traffic analysis techniques and countermeasures	Multiple encrypted traffic datasets	Encryption challenges limit payload inspection; countermeasures are emerging.	Practical deployment under high-speed encrypted flows has not been studied.
Dhiman et al. (2024) [40]	Review of Zero Trust Network Models and Comparative Approaches	Review-based (no dataset)	ZTA is essential for adaptive security; integration strategies vary widely	Mostly conceptual, with limited real-world ZTA deployment studies
Singh et al. (2025) [49]	SHAP-based interpretable anomaly detection in encrypted traffic	Simulated encrypted traffic dataset	SHAP reveals key feature contributions, improving transparency	High computational overhead for SHAP in real-time IDS
Mohale & Obagbuwa (2025) [50]	Systematic review on XAI integration in IDS frameworks	Review-based (various IDS datasets)	XAI enhances IDS trust, but scalability and overhead remain challenges	Gap in lightweight, real-time, explainable IDS solutions

strong anomaly detection, as described in textbooks on monitoring this traffic. Define $D = \{x_i, y_i\}_{i=1}^N$ as dataset of network flow records, where each $x_i \in R^d$ is a d -dimensional vector of statistical features extracted from encrypted flows (e.g., packet count, byte rate, inter-arrival time), and $y_i \in \{0,1\}$ denotes the ground truth label, 0 for benign traffic and 1 for anomalous activity.

The goal is to train a deep neural model defined function $f : R^d \rightarrow [0,1]$ to model the probability of a flow being anomalous as a given flow defined by the model. Minimizing a binary classification loss that is usually taken as the binary cross-entropy function, is the learning objective as in Eq. (1).

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(f(x_i)) + (1 - y_i) \cdot \log(1 - f(x_i))] \quad (1)$$

In a Zero Trust scenario, the detection function works by the assumption that the traffic is encrypted and untrusted by default, so only metadata and statistical features can be used. In addition, extended explainable function f because transparency is essential for security-sensitive environments. Let $\varphi(x_i) \in R^d$ measure of contribution vector: the influence of each input features on the prediction $f(x_i)$. This is done using techniques like SHAP or attention-based attribution mechanisms to calculate $\varphi(x_i)$, leading to human-readable detachments for each detection.

This explains why the anomaly detection pipeline does two things: classifying the flows that are encrypted using the prediction function f , and simultaneously producing also

the explanation vector φ , which elucidates the workings of the model. This helps security analyst to explain their automated decision making and follow up on false positives and undetected threats. This overarching architecture makes trust decisions in Zero Trust Networks data-driven, auditable for compliance and adjusted automatically based on changing behaviors of encrypted traffic.

3.2 Background on deep learning for encrypted traffic

Encrypted traffic on networks poses a significant challenge for existing intrusion detection systems, as it renders payload inspection impossible due to encryption protocols such as TLS, SSH, and VPN. As a result, the focus of research has shifted towards machine learning and deep learning techniques that operate on flow-level metadata rather than raw packet content. As noted above, metadata features – such as byte counts, inter-arrival times, and connection durations – remain visible in encrypted streams and can be used for behavioural pattern recognition.

Deep learning approaches, specifically CNNs and RNNs, demonstrated high-performance modeling for structured flow-based data. While convolutional neural networks (CNNs) can focus on local spatial patterns due to the nature of the sequence of flow statistics, recurrent neural networks (RNN) ability to learn temporal dependencies makes their variants (Long Short-Term Memory (LSTM) for example, Gated Recurrent Units (GRU) and other RNN variants even suitable for learning temporal dependencies within (multiple) flows/session. A GRU layer then takes the sequence as input, updating its hidden state h_t as follows for a sequence of generated flow vectors $\{x_t\}_{t=1}^T$ as expressed in Eq. (2).

$$h_t = GRU(x_t, h_{t-1}) \quad (2)$$

This offers the model temporal context which is crucial for differentiating typical behaviour from the small variations inherent to anomalies in encrypted environments.

Additionally, attention mechanisms have become more popular for weighting time steps or features differently. Let a_t be the attention weight given each of the encoded flow representation h_t , the context vector c can be calculated using Eq. (3).

$$c = \sum_{t=1}^T a_t h_t \quad (3)$$

By doing this, the model learns to concentrate only on the time intervals of a sequence that are most representative of normal vs. abnormal behavior, thus improving detection performance and interpretability.

These deep architectures, if appropriately trained, have been observed to outperform traditional machine learning models by capturing the complex nonlinear dependencies inherent in the encrypted traffic. Because payload content is not available, neural models are suitable for such application domains as their capacity to learn discriminative patterns from structured flow data automatically. Deep learning is, therefore, the backbone of the modern anomaly detectors for Zero Trust networks with encrypted traffic.

3.3 Explainable AI foundations

Deep learning models are slowly gaining traction in the already crowded network security task arena. Still, their lack of transparency is a primary obstacle to their

implementation, especially in essential environments such as Zero Trust Networks. To mitigate this limitation, XAI techniques are fused with anomaly detection frameworks to enable the transparency and interpretability of predictions. Designing XAI aims to make a trained model human-interpretable without compromising performance by making its inner workings understandable to humans.

Given an input x_i , an anomaly score or probability is scalably output by a black-box model $f(x_i)$. Nevertheless, reasoning for this particular choice is key for building trust and usability in security applications. SHapley Additive exPlanations (SHAP), which applies fundamentals from cooperative game theory to provide attribution of each feature's contribution to the final output, is one of the most commonly used techniques. SHAP value $\phi_j(x_i)$ for feature j : Where is a marginal contribution of feature to model output $f(x_i)$ and overall explanation vector is formally defined as in Eq. (4).

$$f(x_i) = \phi_0 + \sum_{j=1}^d \phi_j(x_i) \quad (4)$$

where ϕ_0 is the ground truth model output, and d is the number of features as input. Such an additive decomposition guarantees that every prediction can be explained in terms of its feature contributions, enabling analysts to validate the decision and trace anomalies to the specific behavior.

A second common approach is to incorporate attention mechanisms into the model architecture itself. With the help of attention, the model can weigh the input features or time steps, meaning that it can focus on the input pieces that had the most effect when deciding. The attention a_j power of the feature x_j is multiplied with that feature to form an attention based context vector, as shown in Eq. (3), so that it can be better interpreted concerning what the model learnt to pay attention to.

Such XAI methods unfold twofold functionalities, the first one brings transparency to the black-box predictions of deep learning, and the second one aids human analysts to diagnose false positives or to adjust the trust policies. Explanations as a first-class object in the decision pipeline are integral to ZTXPlainAI, rather than an afterthought, as in the decision pipeline implementation, which aims to address the accountability and transparency aspects of encrypted traffic anomaly detection.

3.4 Zero trust network principles

The Zero Trust Network (ZTN) is a new security paradigm that replaces perimeter-based security models with a new status quo of “never trust, always verify.” A Zero-Trust environment does not trust any entity, inside or outside the organisation, by default. Which means that all access requests, every communication between devices, and all traffic flows need to be dynamically and continuously authenticated, authorised, and validated using adaptable risk assessment. This methodology is critical at the enterprise level, where users and devices are constantly crossing network boundaries and accessing cloud and on-premises resources.

The core concept behind ZTN is to use least privileged access. This principle states that users and devices must not have a higher level of access than needed to complete their task. If traffic is encrypted, it means inspecting all traffic headed to or from the organisation. As encryption prevents access to packet payloads, this sort of anomaly detection must rely on metadata and behavioural signals to infer potential malpractice.

An observed sequence of encrypted traffic flows over time (where each is analyzed beforehand and considered untrusted until proven otherwise) is denoted by $T = \{t_i\}_{i=1}^N$. The decision function $f(t_i)$ (1) translates into a security policy that indicates whether a flow should be allowed, flagged, or quarantined based on the various characteristics of packets and learned threat patterns.

In addition to access control, Zero Trust model's other security solutions to continuously monitor and adjust policy enforcement. The bundling of anomaly detection, behavioural profiling, and real-time trust score updates. This framework is made robust by integrating explainable AI models, where the model does not only predict the anomalies but also explains the reasons behind this prediction to the respective security teams. And those interpretations enable us perform decisions and audits when it comes to high assurance environments, both of which are challenging problems.

As such, this makes the Zero Trust model an ideal strategic framework for how to secure encrypted traffic, and combined with explainable AI, it forms a robust and transparent defense-in-depth strategy that can continuously adapt to changing threats in complex networking environments.

3.5 Notations and symbols

The encrypted traffic anomaly detection framework for Zero Trust Networks consists of several mathematical symbols and notations to define the flow of data, model behavior, and explanation mechanisms of the framework as shown in Eq. The following sections describe key notations used in formulation and model architecture, which allow for clarity and consistency throughout the methodology and analysis.

Let $x_i \in R^d$ be the input feature vector corresponding to the i^{th} encrypted traffic flow given the extracted flow-level features (e.g., byte counts, packet durations, and handshake flags). Ground truth label corresponding to x_i is denoted as $y_i \in \{0,1\}$, where 0 means benign traffic while 1 means anomalous traffic. The dataset is denoted as $D = \{(x_i, y_i)\}_{i=1}^N$, where N is the number of flow instances used in training and evaluation.

When provided with an input vector, the predictive function $f : R^d \rightarrow [0,1]$ produces a score $f(x_i) \in [0,1]$, which is interpreted as the likelihood of the input vector being anomalous. We parameterize this model using a deep neural architecture based on layers including convolutional encoders, recurrent units, and attention blocks. In the time-sequenced flows, the hidden state of the recurrent layer at time t is denoted by h_t , and the attention weight h_t of is represented by a_t . As before, the aggregated context vector is computed as $c = \sum_{t=1}^T a_t h_t$, as defined in Eq. (3).

In the explainability module, contribution of feature j to the output for the instance x_i is denoted by $\phi_j(x_i)$ where $f(x_i)$ is the SHAP value associated with j^{th} feature. Finally, the contributions are added to the base value ϕ_0 to reconstruct the model prediction as $f(x_i) = \phi_0 + \sum_{j=1}^d \phi_j(x_i)$, in Eq. (4). These symbols underlie the mathematics behind the ZTXPlainaAI framework and are used throughout the methodology and results for consistent interpretation and reproducibility of the proposed system. Table 2 presents the key mathematical symbols and notations defining datasets, model components, parameters, and explainability elements in ZTXPlainaAI.

Table 2 Notations and symbols in ZTXPlainAI framework

Symbol	Description
$x_i \in R^d$	Feature vector for the i^{th} encrypted traffic flow with d dimensions
$y_i \in \{0,1\}$	Ground truth label: 0 for benign flow, 1 for anomalous flow
D	Full dataset comprising flow-feature and label pairs
$f(x_i)$	Anomaly score output from the EncXplainNet model for input x_i
$\phi_j(x_i)$	SHAP value representing the contribution of feature j in x_i
ϕ_0	Base value (expected output) in SHAP decomposition
z_i	Feature representation after CNN transformation
h_t	Hidden state of the GRU at time step t
a_t	Attention weight assigned to hidden state h_t
c	Context vector computed from attention-weighted hidden states
θ	Model parameters before feedback-based update
θ'	Updated model parameters after incorporating analyst feedback
D_{train}	Original training dataset
D_{feedback}	Analyst-verified feedback dataset
L	Loss function during training
L_{total}	Combined loss over training and feedback data
η	Learning rate used in optimization
m_t, v_t	First and second moment estimates in Adam optimizer

4 Materials and methods

In this section, the proposed ZTXPlainAI framework is developed and evaluated. It describes the dataset preparation and preprocessing, system overview and architecture of the EncXplainNet model. It also describes the integration of explainability, a reinforcement feedback loop, training setups, evaluation procedures, and threat models to serve as a basis for experimental analysis.

4.1 Dataset and preprocessing

The current approach utilises the CIC-IDS2019 dataset, a recent large-scale benchmark dataset for intrusion detection from the Canadian Institute for Cybersecurity. ML Pro-Gent dataset. It offers near-real network traffic, along with reference data for different protocols (HTTPS, SSH, and VPN), including benign and attack data. This becomes especially useful when considering zero-trust anomaly detection use cases, including flows with encrypted traffic and limited visibility into the payload. This facilitates supervised learning, as these traffic sessions have their class (benign or attack type) known.

To obtain network flows from raw PCAP files, the open-source tool CICFlow Meter aggregates packets into bidirectional flows using common five-tuple (source IP, destination IP, source port, destination port, and protocol) criteria. We summarize each flow into a vector of statistical features, including features like flow duration, total bytes sent and received, average packet size, inter-arrival times, TCP Flags counts, etc. Let us denote each flow by the feature vector $x_i \in R^d$, where d is the number of extracted features and x_i represents the i^{th} flow from the data dissected.

Each feature is normalized, using min-max scaling, to improve the quality of learning and enforce numerical stability. For the feature dimension j , we compute the normalized value $\hat{x}_{i,j}$ as in Eq. (5).

$$\hat{x}_{i,j} = \frac{x_{i,j} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (5)$$

where $\min(x_j)$ and $\max(x_j)$ represent the minimum and maximum of the j^{th} feature, respectively, over the training set. This scaling forces all features to fall within the $[0,1]$ range, thereby ensuring compatibility with activation functions and accelerating convergence during the training phase.

The aggregated dataset is then pre-processed and divided into training, validation, and test sets through stratified sampling to preserve the class balance over partitions. Also, part of encrypted flows (such as HTTPS and SSH traffic) is intentionally kept to emulate encrypted conditions of the environment of Zero Trust Network. Feature-label pairs $\{(x_i, y_i)\}_{i=1}^N$ are the objects defined in Sect. 3.1 as input to the model to be trained and evaluated. Such a preprocessing pipeline ensures that the anomaly detection model operates on encrypted (and thus unidentifiable) metadata, aligning perfectly with the privacy-preserving nature of Zero Trust architectures.

4.2 System overview: ZTXPlainaAI

The proposed ZTXPlainaAI framework, presented in Fig. 1, enables explainable deep-learning techniques for anomaly detection in encrypted traffic flows of Zero Trust Network environments. And its premise is that no traffic can be trusted by its source or destination, so each encrypted session, internal or external, must be inspected and validated. This framework combines data-driven inference with explainable decision-making, thus providing security operations with accuracy and interpretability.

The system first ingests the live or previously stored encrypted traffic (only if the security analysis policy is to protect proactively) from any source, such as enterprise devices, VPN clients, and IoT nodes. Those traffic streams are all processed through a flow extraction engine, which is responsible for transforming raw information about packets into bidirectional flow records. The first part of the flow is converted to a feature vector—where a feature vector contains structural features $x_i \in R^d$ such as number of

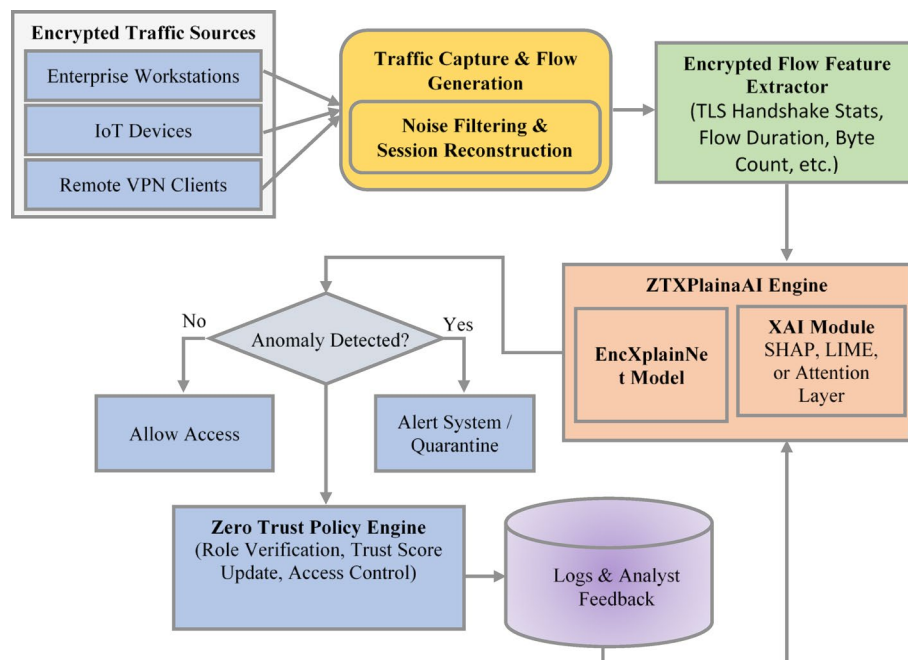


Fig. 1 System architecture of ZTXPlainaAI for encrypted traffic anomaly detection

packets, number of bytes, flow duration, and flags at protocol level. We use them as the input features for the anomaly detection engine.

The heart of the framework is EncXplainNet, a hybrid deep learning model that takes the extracted features as input and predicts the anomaly probabilities. The model then returns, for every input x_i , a score $f(x_i) \in [0,1]$, representing the probability that this traffic flow is an anomaly. This score is subsequently compared against a detection threshold to decide if the flow should be reported for additional verification or it should be let through.

To improve decision traceability, ZTXPlainAI include an explainability module that can generate a feature attribution vector $\varphi(x_i)$, corresponding to the input. The purpose of this explanation is to show which features most impacted the prediction and to allow security analysts to understand the rationale behind each decision. In Zero Trust Networks interpretability is a must-have because policy enforcement must be auditable and based on evidence.

If a flow is deemed anomalous, it will be sent to the Zero Trust policy engine for enforcement actions like access denial, quarantine, or multi-factor re-authentication. At the same time, the system logs the explanation vector and decision outcome, allowing a human analyst to review them simultaneously. In subsequent training cycles, analyst-oriented reports can be used to adjust the model, contributing to a shaping loop that increases detection precision over time.

As shown in Fig. 1, the system architecture comprises this end-to-end chain of the pipeline from encrypted traffic ingestions, anomaly detection, explanation generation, policy enforcement, and continuous model adaptation to support a resilient and interpretable defense mechanism for Zero Trust environments.

Figure 2 depicts a processing chain for how encrypted traffic can be ingested and analyzed in order to detect anomalies, explain those anomalies, enforce policy and provide feedback to the analyst. For each stage, encrypted flow candidate data are first extracted and preprocessed for feature vector construction. These feature vectors are processed by the EncXplainNet model and are used to calculate the anomaly scores along with explanations in form of SHAP or attentions. Using these findings, Zero Trust policies determine whether to permit, deny or quarantine traffic flows. At the same time, explanations and decisions are logged and shown to analysts, whose feedback goes in the retraining loop for repeatedly improving models and adapting network security.

4.3 Model architecture: EncXplainNet

ZTXPlainAI incorporates an enhanced version of a deep learning model for anomaly detection, named EncXplainNet, as the key engine that can be utilized for finding anomalies in the encrypted traffic flows, using a hybrid architecture for balancing predictive accuracy against the level of interpretability. Due to the absence of visibility to the payload of encrypted network traffic, the model thus only relies on flow-level statistical features outputs during pre-processing, as described in Sect. 4.1. The model is fed with a normalized feature vector $x_i \in R^d$ as input where d is the number of flow features corresponding to a single encrypted session.

Figure 3 depicts the EncXplainNet model, integrating CNN, GRU, attention, and SHAP-based explainability for encrypted traffic anomaly detection. EncXplainNet uses a sequential configuration of convolutional and recurrent layers to capture local feature

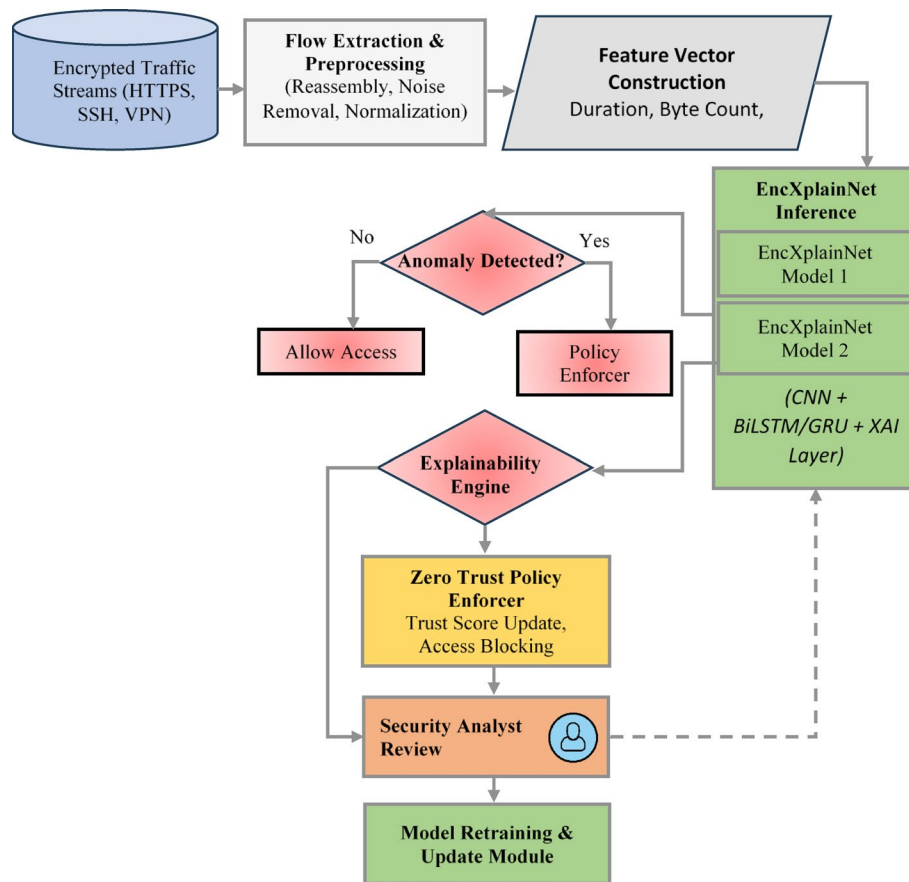


Fig. 2 Workflow of the ZTXplainAI framework

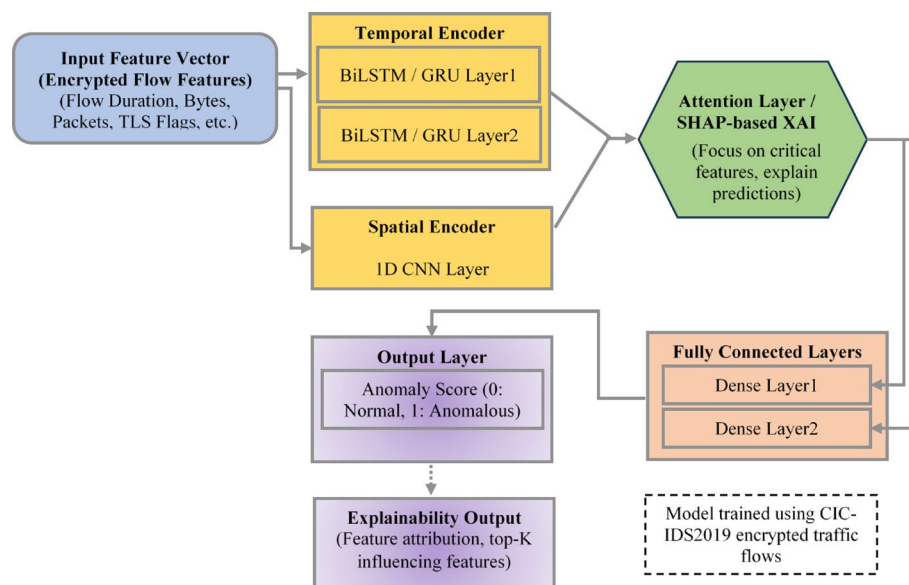


Fig. 3 Model architecture of EncXplainNet for encrypted traffic anomaly detection with integrated explainability

patterns and long range temporal dependencies effectively. First, we reshape the input feature vector x_i into a 1D tensor for doing convolutional riddance. Next, we apply a spatial feature extractor using a 1D CNN layer that expects to extract high-level spatial features z_i that depict the relationship between neighboring statistical features as in Eq. (6).

$$z_i = \text{ReLU}(\text{Conv1D}(x_i)) \quad (6)$$

The intermediate representations z_i then go through a Gated Recurrent Unit (GRU) layer for learning any temporal or sequential relationships between them, which is especially beneficial when network traffic flows belong to a session that comprises of many packets or within multi-step sequences of attacks. Some parts of the GRU updates its hidden state h_t at each time step t as outlined in Eq. (2), which we have already discussed.

For interpretability, we add an attention mechanism over the GRU outputs so as to make the model pay attention to the most relevant features. More formally, the attention layer calculates a weighted context vector c by scoring each hidden state with a corresponding importance score a_t , as described in Eq. (3). This vector summarizes the most informative dimensions of the traffic flow and is used as input to the classification head.

After the attention mechanism, the context vector c is passed to a series of fully connected (dense) layers which successively reduce dimensionality and project features for binary classification. Output layer applies sigmoid activation and outputs an anomaly score $f(x_i) \in [0,1]$, indicating the probability that the input flow is anomalous.

EncXplainNet is also integrated with SHAP-based post hoc explanation methods to support explanation. The SHAP algorithm computes for each prediction $f(x_i)$, a vector of feature contributions $\phi(x_i) = \{\phi_1(x_i), \phi_2(x_i), \dots, \phi_d(x_i)\}$, such that the additive decomposition in Eq. (4) is satisfied. That explanation vector breaks down the contribution of each input feature towards the final decision in a way that humans can understand.

In EncXplainNet, local feature interactions are first captured using a one-dimensional convolutional layer with a kernel size of 3 and a stride of 1, followed by ReLU activation. Temporal dependencies across encrypted traffic flows are modeled using two stacked GRU layers, whose hidden states are processed by an additive attention mechanism to emphasize informative temporal segments. The resulting attended representation is passed through two fully connected layers with 64 and 32 neurons, respectively, each employing ReLU activation and dropout regularization applied after the dense layers. A final sigmoid-activated output layer produces the anomaly probability, completing the end-to-end detection pipeline.

EncXplainNet hybridizes convolutional encoders, recurrent processors, attention-based summarization, and XAI, thus permitting it to both detect subtle anomalies in encrypted flows and explain its predictions in a way ready for analyst evaluation and policy updating.

4.4 Explainability integration

The role of explainability in ZTXPlainaAI is pivotal, as evidenced by the case where not only is the anomaly detection decision in the EncXplainNet model taken correctly, but it is also interpretable and auditable. Justifications for detection output: Since deep

learning models are said to be too opaque to understand, and because decisions made within Zero Trust Network environments can be critical, it is essential to know why a particular output was detected. This framework links explainability in two ways: internal interpretability (attention-based) and post hoc attribution (using SHAP).

EncXplainNet model has an attention mechanism within itself to compute the importance weights a_t dynamically to each hidden state h_t generated at sequential feature processing to identify the attributions of the encoder inputs. The mechanism enables the model to pay closer attention to certain time windows or characteristics of the encrypted flow being more relevant for abnormal behavior. They then compute a context vector c using the attention weights Eq. (3), combining the most relevant information for final decision-making. By plotting the values of a_t we can see which time steps or flow characteristics influenced the output the most, providing an underlying level of interpretability straight out of the box.

In addition to this intrinsic mechanism, the framework incorporates SHapley Additive exPlanations (SHAP) as a model-agnostic post hoc interpretability method. For a single prediction, SHAP calculates a set of values $\phi_j(x_i)$, for each input feature vector x_i , characterizing the contribution of feature j to prediction $f(x_i)$. This means that the additive property presented in Eq. (4) is satisfied and thus, the final model prediction can be explained as the sum of the individual feature contributions and a base value ϕ_0 . This makes it clear and mathematically based reasoning for each prediction, allowing the analyst to see whether flow duration, packet inter-arrival time or byte count were the causative features which flagged a traffic flow as anomalous.

In this case, the SHAP explanations for the predicted anomaly across all samples and the classification output are logged together. The explanations are then visualised through interactive plots (i.e., bar charts, waterfall plots, ranked lists of features) for further investigation by the security analyst. In essence, this augments confidence in the system whilst also helping refine Zero Trust access policies based on the user behaviour attributes the model classifies as malicious or benign.

Together, this attention layer and SHAP operate as a dual-explanation pipeline, with one providing real-time, internal interpretability information about the model and the other generating externally verifiable explanations in the form of SHAP features for security auditing and analyst-driven investigation. This integration fortifies the ZTXPlainaAI framework, making it robust, lucid, and deployable in situations where accuracy and responsible action are paramount.

4.5 Reinforcement and feedback loop

ZTXPlainaAI: Added a feedback loop to enable continuous learning and adaptation in ever-changing Zero Trust Network environments. The encrypted traffic patterns are variable, and the attack strategies are adapting, thus a static anomaly detection model might lose its detection capability with time. To mitigate this problem, the framework facilitates a human-in-the-loop feedback mechanism by allowing security analysts (e.g., practitioners using our framework) to review model outputs, confirm predictions, or provide corrective feedback to improve the model.

While processing a flow, the EncXplainNet model outputs both a prediction $f(x_i)$ and an explanation vector $\phi(x_i)$, and these results along with metadata (timestamp, session id, policy action, etc.) are stored into a log buffer. This is provided to the security

analyst in the form of a dashboard interface through which the analyst can examine detected anomalies, validate ground truth labels, and the interpretability outputs produced by either SHAP or attention mechanisms.

If the analyst does not agree with the prediction of the model — namely a false positive or confirming a missed anomaly — the label y'_i is stored with the feature vector x_i . This collection of manually validated instances makes up a supporting dataset $D_{\text{feedback}} = \{(x_i, y'_i)\}$ which is utilized for scheduled model retraining or incremental updates.

Consider a base model denoted by θ , and a new model following integration of feedback represented by θ' . Lets define the new loss function over the joint training set $D_{\text{train}} \cup D_{\text{feedback}}$ as expressed in Eq. (7).

$$L_{\text{total}} = -\frac{1}{|D|} \sum_{(x_i, y_i) \in D} [y_i \log f_{\theta'}(x_i) + (1 - y_i) \log(1 - f_{\theta'}(x_i))] \quad (7)$$

Where $D = D_{\text{train}} \cup D_{\text{feedback}}$, and $f_{\theta'}(x_i)$ is the prediction of the updated model after fine-tuning. This stepwise improvement enables the model to specialize based on analyst-led supervision while retaining knowledge from earlier training.

Simultaneously, information articulating analyst validation is utilized to shape Zero Trust policy rules incrementally. When a benign nuisance pattern occurs repeatedly but is falsely identified as anomalous, one can adjust the matching threshold or condition to be less conservative, and thereby minimize false positives. On the other hand, patterns that have been consistently associated with any known anomaly can be automated to create a rule template or used to design an artificial attack instance to enhance the robustness of the model.

As such, this feedback loop reinforces the role of machines and humans → in the detection cycle such that ZTXPlainAI is a self-healing, audit-able system. The provision of explainability during both stages of detection and feedback offers the trust, adaptability and accountability that one would expect from deployment under the concepts of a Zero Trust Network.

Leveraging analyst-guided feedback as a human-in-the-loop design component that allows selective retraining on few samples verified by an analyst based on evolving threat conditions. This mechanism is not tested as a stand-alone adaptive learning mechanism in the current study, nor compared against non-adaptive baselines. Instead, it is more a proof-of-concept integration to demonstrate the workflow of how human feedback can be integrated into the model lifecycle in Zero Trust environments.

4.6 Algorithmic implementation

Here we demonstrate algorithmic implementation of the ZTXPlainAI framework as well as its generic model, EncXplainNet. We describe the protocols for performing anomaly detection, integration of explainability, and adaptive feedback, step by step. Results demonstrate that the algorithms capture spatial-temporal flow patterns, provide transparent feature attributions, and allow the analyst to iteratively validate key components, leading to high detection accuracy in Zero Trust environments while maintaining interpretability.

The usage of the EncXplainNet model to detect anomalies in EncXplainNet analysis of encrypted traffic flows is illustrated in Algorithm 1. Output is a single traffic flow (in

Algorithm: Flow-Based Anomaly Detection Using EncXplainNet

Input: Encrypted flow feature vector $x_i \in \mathbb{R}^d$

Output: Prediction $\hat{y}_i$, Explanation $\phi(x_i)$

- 1: Normalize x_i using min-max scaling
- 2: Reshape x_i for 1D convolution input
- 3: Compute $z_i = \text{ReLU}(\text{Conv1D}(x_i))$
- 4: Pass z_i through GRU to obtain hidden states $h_1, h_2, \dots, h_T$
- 5: Compute attention weights a_t for each hidden state h_t
- 6: Compute context vector

$$c = \sum_{t=1}^T a_t h_t$$
- 7: Feed c into fully connected layers
- 8: Compute anomaly score

$$\hat{y}_i = f(x_i) \in [0,1]$$
- 9: If $\hat{y}_i > \tau$ then
 Label flow as anomalous
- 10: Generate explanation vector

$$\phi(x_i) = \{\phi_1, \phi_2, \dots, \phi_d\}$$
 using SHAP
- 11: Output prediction $\hat{y}_i$ and explanation $\phi(x_i)$

Algorithm 1 Flow-based anomaly detection using EncXplainNet

Algorithm: Analyst-Guided Feedback Integration for Model Refinement

- 1: Input: Trained model f_θ , analyst-labeled feedback set $D_{\text{feedback}} = \{(x_i, y'_i)\}$
- 2: Merge feedback with training data: $D = D_{\text{train}} \cup D_{\text{feedback}}$
- 3: Initialize updated model parameters $\theta' \leftarrow \theta$
- 4: For each training epoch do
- 5: Shuffle D and create mini-batches
- 6: For each mini-batch $B = \{(x_j, y_j)\}$ do
- 7: Compute predictions $\hat{y}_j = f_{\theta'}(x_j)$
- 8: Compute loss L_{total} using Equation (7)
- 9: Update parameters θ' using Adam optimizer
- 10: Output: Updated model $f_{\theta'}$

Algorithm 2 Analyst-guided feedback integration for model refinement

statistical feature vector $x_i \in \mathbb{R}^d$) as input to the Algorithm. Binary output as traffic flow of either it is malicious or normal. Step 2 Normalizes the vector using min-max scaling, such that all features belong to a specific range of numbers. This helps in model convergence and stability Step 3: Reshape the data as per the input dimensions required by the 1D convolutional layer.

In step 4, it uses a 1D convolution and a ReLU activation to learn the spatial correlations of neighboring features. Step 5 then feeds the representations through an n-grams Gated Recurrent Unit (GRU) layer, which models sequential dependencies and maintains contextual patterns seen across flow features. In step 6, attention weights are calculated over the hidden states, so relative importance of each of the encoded representations can be decided. The context vector c , used as an input in Step 7, is a weighted sum of the hidden states and serves to summarize the most salient properties of the flow.

In step 8, this context vector is passed through two fully connected layers to produce a binary decision on whether context expresses the meaning of a tweet or not. Step 9: The model generates a probability score $\hat{y}_i = f(x_i) \in [0,1]$, which represents the probability that this flow is anomalous. If the flow is classified as benign or anomaly in step 10, as shown in Fig. 4, we apply a threshold to determine the output flow type. Step 11: SHAP we use here to calculate the explanation vector $\phi(x_i)$, which measure each input feature contribution to the final prediction. The algorithm provides the prediction along with its explanation, facilitating the interpretable decision-making in Zero Trust.

Algorithm 2 focuses on the reinforcement mechanism that refines the EncXplainNet model with the feedback of the security analysts. We define an algorithm that starts from a trained model f_θ and a feedback dataset D_{feedback} with the encrypted traffic samples x_i and labels y'_i approved by a reviewer after reviewing model predictions and explanations.

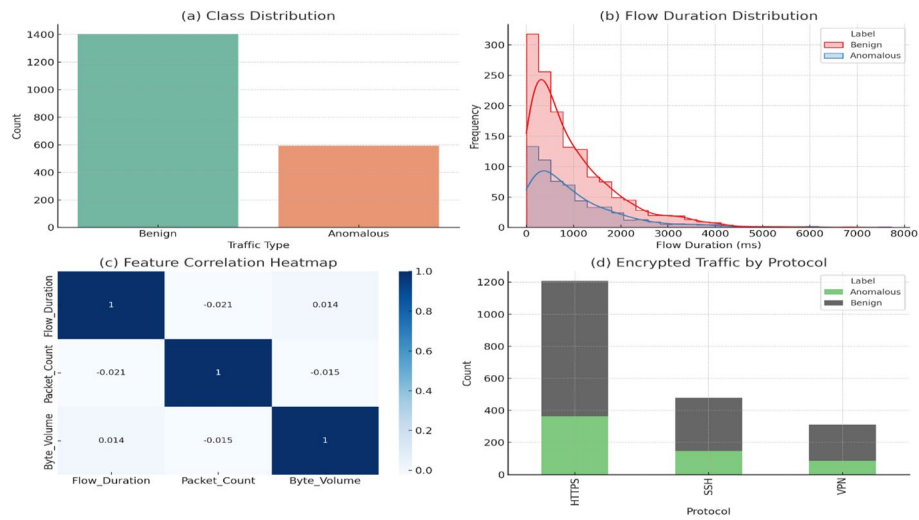


Fig. 4 Exploratory data analysis of CIC-IDS2019 encrypted subset showing class distribution, flow duration, feature correlations, and protocol-wise traffic

Step 2 combines the original training dataset D_{train} with the feedback samples to get an augmented dataset D . This ensures that while the model can learn from newly identified false positives or false negatives, it also preserves what it knew earlier. We initialize the model parameters θ' from the original parameters θ of the two-branch architecture, in order to fine-tune instead of training from scratch.

We have points 4 to 9 for the mini-batch training loop. In each epoch, the dataset containing both the real and the synthetic data array will be shuffled and batched. We compute the predictions for each mini-batch, evaluate loss L_{total} using the updated binary cross-entropy loss from Eq. (7). Finally, we update the parameters θ' with the Adam optimizer that adapts the learning rate and is recommended for stabilizing convergence.

Repeating this process for several epochs allows the model to adjust cumulatively to new traffic and use the adjusted labels. The end result is the model $f_{\theta'}$, which has been refined with other humans to correct what you now know is likely countless misclassifications of infections so that (hopefully) in the future we minimize further detections of infections that would otherwise be misclassified. This algorithm supports continual learning, ensuring ZTXPlainaAI remains effective even in a Zero Trust environment where attackers' behavior and encrypted traffic evolve regularly.

4.7 Threat model and assumptions

ZTXPlainaAI has a threat model based on a Zero Trust Network architecture, which does not trust any internal or external network/traffic. The system assumes that attackers have already breached the network perimeter, which is protected by stronger encryption channels against suspicious and malicious activity. Thus, the framework here is specifically oriented towards advanced threats, where target traffic flows are encrypted and so would circumvent standard signature or payload inspection approaches.

The primary threat this model targets is a network-level attacker utilising encrypted communication protocols, such as HTTPS, SSH, or VPN, for reconnaissance, lateral movement, C2, data exfiltration, or DDoS attacks. These threats are typically hidden in flows whose packets appear normal but can behave abnormally due to timing, volume, or statistical properties. We do not decrypt the payload either – such a capability is

neither practical nor desirable for privacy and computational reasons – thus, the rest of the model operates only on the available flow-level metadata.

Although adversaries may attempt to shape encrypted flows to resemble benign traffic, we do not assume that flow-level statistics are immutable. In particular, sophisticated attackers can alter timing, burstiness, packet sizes, and byte fractions while preserving protocol semantics using traffic morphing, adaptive padding/fragmentation, or rate control. Accordingly, our threat model considers an adaptive evasion adversary that can partially manipulate observable flow metadata under operational constraints (e.g., latency overhead, bandwidth cost, and throughput degradation). Under this setting, ZTXPlainaAI is intended to detect residual behavioral inconsistencies and temporal irregularities that arise when attackers emulate benign baselines at scale. Evaluating robustness against fully optimized traffic-morphing strategies is a non-trivial extension and is identified as a limitation and future work in Sect. 6.

The defender has access to encrypted traffic flow metadata, system and policy logs, historical labeled datasets, and analyst feedback. An assumption is also made that the ecosystem allows protection mechanisms, viz. detection models and policies, to be updated regularly and incrementally, using real-time threat intelligence. Even though the framework runs initially in a supervised manner with data that has been prepared and labelled, it has a feedback loop (as indicated in Sect. 4.5) that can be used to adjust and improvements over time.

The system does not assume knowledge of specific attack types in advance; instead, it is designed to generalize patterns of abnormality across diverse encrypted traffic scenarios. Additionally, the model assumes that encrypted flows are not tampered with in transit (i.e., the encryption is preserved), and that flow records are accurately collected without packet loss or sampling errors that could distort feature distributions.

In closing, the threat-model is overly pessimistic and Zero Trust, but it is a reminder that encrypted flows from both inside and outside of your network must be ceaselessly monitored, sampled and accounted for. These are enterprise-scale network deployments, leading to realistic assumptions guiding the design of the EncXplainNet detection engine and its supporting framework ZTXPlainaAI.

5 Experimental results

In this section, we describe the experimental evaluation of the ZTXPlainaAI framework and EncXplainNet model on encrypted traffic anomaly detection. They demonstrate the performance of the proposed approach compared with classical and deep learning baselines using the CIC-IDS2019 Dataset. One way to exhibit accuracy, interpretability, and adaptability in the context of ZTs is to provide performance results, insights into explanation and robustness tests, and comparative performance benchmarks with the SoA.

5.1 Dataset description and preparation

Experiments were conducted using the CIC-IDS2019 dataset [81], provided by the Canadian Institute for Cybersecurity, which comprises enterprise-level network traffic spanning multiple days and includes both benign user activities and a range of attack scenarios. In this work, we consider only the subset of encrypted traffic, including flows from protocols such as HTTPS and SSH, for anomaly detection under Zero Trust.

Raw packet captures were transformed into bidirectional flow records using CICFlowMeter, which correlates packets based on the standard five-tuple (source IP, destination IP, source port, destination port, and protocol). Each flow was characterised as a feature vector of session-level attributes, including flow duration, total numbers of forward and backward bytes, packet counts (total and in each direction), statistics on packet inter-arrival times, and the presence or absence of local TCP flags. Payload-related characteristics have been intentionally omitted for the sake of encryption privacy, resulting in feature vectors $x_i \in \mathbb{R}^d$, where d denotes the number of extracted flow-level attributes.

This yielded a total of 2,000 flows in the encrypted traffic subset, of which 1,400 (70%) were benign and 600 (30%) were anomalous after pre-processing. While this distribution is much more moderate than the severe imbalance typically seen in production environments, it still introduces an inherent skew toward benign traffic in anomaly detection tasks. If not properly addressed, this distribution can lead to biased learning for the majority class. To reduce this effect during training, class weights were calculated inversely proportional to class frequencies, with misclassified anomalous flow receiving a relatively higher penalty. This acts the same way, improving the rate of picking minority attack instances during the learning process while preserving learning equilibrium among classes.

To minimise bias arising from differing feature scales and enhance convergence stability, all numerical features were normalised using min–max scaling Eq. (5), mapping values to the range $[0, 1]$. Some attributes, such as protocol identifiers, were converted to numerical values because they were categorical and deep learning models accept only numerical inputs.

Lastly, the dataset was split into three folds using stratified sampling, keeping the class distribution the same as the original dataset: training (70% flows), validation, and test (each with 15% flows). This approach allowed for a fair evaluation without compromising the class proportions. In Zero Trust environments with extreme class imbalance, transparent reporting of performance metrics is only possible by explicitly reporting the class distribution and class weights during preparation.

Even though the dataset used in this study does not exhibit the significant imbalance typically found in a large-scale operational network, anomaly detection is still naturally biased toward normal traffic. Typically, benign flows outnumber malignant ones in practical deployments; therefore, class-sensitive evaluation metrics such as precision, recall, F1-score, and PR-AUC are the natural choice. To learn effectively under this mild class imbalance, we used class-weighted loss functions during training, so that misclassifying an anomalous flow incurs a higher penalty.

Figure 4 illustrates exploratory data analysis of the CIC-IDS2019 encrypted subset. Subfigure (a) shows the distribution of benign versus anomalous traffic. Subfigure (b) presents flow duration distributions across classes. Subfigure (c) highlights feature correlations among key flow attributes. Subfigure (d) depicts protocol-wise traffic patterns, comparing the proportions of benign and anomalous traffic across HTTPS, SSH, and VPN sessions.

Figure 5 presents additional exploratory data analysis of the CIC-IDS2019 encrypted subset. Subfigure (a) illustrates cumulative traffic volume over time, highlighting temporal fluctuations. Subfigure (b) ranks features by variance-based importance. Subfigure (c) shows PCA projection, revealing the separability of benign and anomalous flows.



Fig. 5 Exploratory data analysis of CIC-IDS2019 encrypted subset showing temporal patterns, feature importance, PCA projection, and byte volume distribution

Subfigure (d) compares byte volume distributions across traffic classes, emphasizing differences between benign and anomalous sessions.

5.2 Experimental setup

To validate research findings, it is very important to perform replicability, and therefore all the experiments were performed in the same computing environment in order to provide a fair evaluation of the proposed frameworks. Workstation with 3.6 GHz Intel Core i9, 64 GB RAM and 24 GB of NVIDIA RTX 3090 While the auto-like system does well for GPU training, the actual system runs Ubuntu 22.04 LTS for stability, and improved driver support.

We configured the software stack whose primary programming language was Python 3.9. TensorFlow 2.11 and Keras were used for the deep learning models, as well as the EncXplainNet architecture, and PyTorch 1.13 for experimenting on the interpretability modules. Libraries such as Scikit-learn, NumPy, and Pandas were used to preprocess the data, whereas SHAP and LIME were used for explainability analysis. The visualisation tasks were completed using Matplotlib and Seaborn.

To accelerate convergence, the EncXplainNet model was trained on a GPU. We used the Adam optimiser with $\eta = 0.001$ as the learning rate and the model's binary cross-entropy loss defined in (1). A minibatch size of 64 and training for up to 50 epochs with early stopping (no improvement in validation loss for 5 epochs) were used. Furthermore, at the fully connected layers, a dropout rate of 0.3 was used to mitigate overfitting, and class weights were applied to address an imbalance between benign and anomalous samples.

The model's hyperparameters were selected through exploratory tuning on the validation set. The architecture used a single causal 1D convolutional layer with a kernel size of 64, three filters, a GRU layer with 128 hidden units, and an attention mechanism to compute weighted context representations. The dense classification head included two dense layers with sizes 64 and 32, respectively, and a sigmoid output neuron to produce

anomaly scores. Optimisation settings and hyperparameters kept training dynamics stable, quick convergence, and generalisation on the encrypted traffic subset of CIC-(IDS2019).

Ten independent runs, varying the random initialisation and data shuffling, were performed for each experiment to ensure statistical robustness. The reported performance figures are the mean values from these runs, while the variability of the results is examined using standard deviation and boxplots.

5.3 Baseline models for comparison

To robustly validate the performance of the proposed EncXplainNet model, several baseline methods were developed and compared across categories, including traditional machine learning, deep learning, explainability-agnostic, and explainable methods. This maintained a fair benchmark and emphasised the potential advantages of interpretability for encrypted-traffic anomaly detection.

They have been used in intrusion detection since the early days, including in flow-based anomaly detection, which is why we opted for traditional machine learning classifiers. Random Forest (RF) was used with 200 trees and Gini impurity as the splitting criterion, leveraging its robustness and ability to capture nonlinear interactions among features. For our more complex margin-based classifier, we used the Support Vector Machine (SVM) with an RBF kernel, since this kernel scales to high-dimensional feature spaces, and we also used Logistic Regression as a linear baseline classifier to compare against our simpler decision boundaries.

Deep learning baselines were included to compare EncXplainNet with existing neural architectures applied to encrypted traffic analysis. A standard 1D Convolutional Neural Network (CNN) with two convolutional layers and max pooling was trained to capture local spatial dependencies among flow features. A Long Short-Term Memory (LSTM) network with 128 hidden units was used to model temporal relationships in sequential flows. At the same time, a stacked Autoencoder was employed as an unsupervised baseline for anomaly detection by reconstructing flow features and flagging instances with high reconstruction error. These models were optimised in the same training environment and hyperparameter ranges as described in Sect. 5.2.

We differentiated between models that are inherently not interpretable and those (such as several example techniques mentioned above) in which explainability is built in through integration with explainable AI methods for analysis. The CNN, LSTM, and Autoencoder baselines were explainability-agnostic, yielding predictions absent feature-level attributions. On the other hand, EncXpla and inNet combined an intrinsic attention mechanism with a post hoc SHAP analysis to produce the explanation. Furthermore, for comparison, the feature importance scores from Random Forest were incorporated into the model, and a simple explainable LSTM with an attention layer was implemented.

Such a comparative set of baselines ensured that EncXplainNet was compared against both traditional classifiers and deep learning models, with a focus on interpretability. Leveraging knowledge of the attack paths, the evaluation framework demonstrates the merits of using predictive performance and explanation quality to deploy ZTXPlainAI in Zero Trust Network environments, where decision transparency is as important as detection accuracy.

5.4 Performance evaluation metrics

Various classification metrics were used to evaluate the performance of the proposed EncXplainNet for anomaly detection in encrypted traffic flows. To begin with, we calculated Accuracy to measure the overall ratio of correctly classified flows (benign and anomalous). Nevertheless, in intrusion detection tasks accuracy alone may give a misguided understanding due to the nature of network level datasets that are characterized by class imbalance, with the presence of a class of flows that are beneficial, compared to a second class that are abnormal.

To compensate for these limitations, additional metrics were incorporated, including precision, recall, and F1-score. Precision reflects the model's ability to minimise false alarms, since it measures the fraction of all flows predicted as anomalous that are actually anomalous. Recall is the percentage of total anomalous flows correctly detected, indicating how sensitive the system is to potential threats. The F1-score is the harmonic mean of precision (the ratio of relevant instances among retrieved instances) and recall (the ratio of relevant instances among actually relevant instances), thus providing a symmetric measure.

The class imbalance in encrypted traffic anomaly detection warrants further utilisation of Precision–Recall (PR) curves and the PR-AUC metric. In particular, PR-based evaluation is informative for minority-class detection, as it emphasises the precision-recall trade-off across different decision thresholds. To enable clearer interpretation of anomaly-detection performance under imbalanced conditions, we plotted PR curves for both EncXplainNet and all baseline models in this study.

Second, the Area Under the Receiver Operating Characteristic Curve (AUC) was employed to assess the model's discriminatory power at a range of decision thresholds. Using AUC as a threshold-independent metric, we can ensure a fair comparison between EncXplainNet and the baseline approaches. In addition, FPR (False Positive Rate) and FNR (False Negative Rate) were calculated to evaluate the practical implications of detection errors. The FPR is the ratio of innocent flows wrongly labelled as anomalous, which is important in Zero Trust environments because too many false alarms can disrupt regular network operations. On the other hand, FNR refers to the percentage of abnormal flows incorrectly categorised as benign, and it is an indicator of missed threats.

5.5 Results on anomaly detection performance

In this subsection, we compare the anomaly detection performance of the proposed EncXplainNet model against several baseline approaches. We report multiple quantitative performance measures (accuracy, precision, recall, F1-score, AUC and PR-AUC), and provide both tabular summaries and graphical visualisations. A crucial PR analysis is included here to address the significance of false positives and false negatives in an anomaly detection task applied to class-imbalanced data, given that correctly classified instances of minority attack classes are of utmost importance. The comparative results show that, regardless of the evaluation metric, EncXplainNet consistently outperforms the baseline models. The results are statistically significant, demonstrating the architecture's ability to detect anomalous encrypted traffic whilst offering strong precision–recall trade-offs in Zero Trust environments.

Table 3 Comparative performance of encxpainnet and baseline models on encrypted traffic anomaly detection (results represent mean performance across 10 independent runs, with variability analysed using standard deviation and boxplot distributions)

Model	Accuracy	Precision	Recall	F1	AUC	PR-AUC
Logistic regression	0.87	0.84	0.82	0.83	0.90	0.86
SVM	0.90	0.88	0.87	0.87	0.92	0.89
Random forest	0.92	0.90	0.89	0.89	0.94	0.91
CNN	0.91	0.89	0.88	0.88	0.93	0.90
LSTM	0.92	0.91	0.90	0.90	0.94	0.92
Autoencoder	0.88	0.85	0.83	0.84	0.91	0.87
EncXplainNet	0.96	0.95	0.96	0.96	0.98	0.97

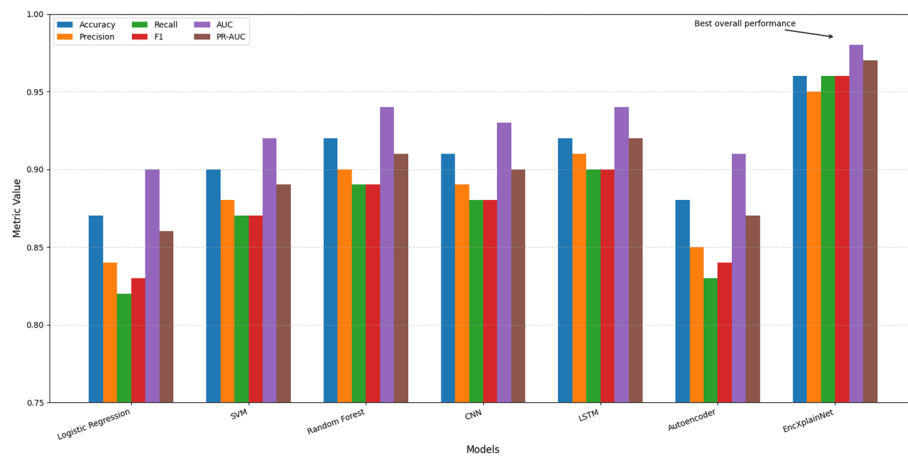


Fig. 6 Comparative multi-metric performance of EncXplainNet and baseline models for encrypted traffic anomaly detection in Zero Trust Networks

The comparison of EncXplainNet with traditional machine-learning and deep-learning baseline models for encrypted-traffic anomaly detection is shown in Table 3. This includes measuring performance in terms of accuracy, precision, recall, F1 score, AUC, and PR-AUC. Results show that EncXplainNet outperforms the baselines across all metrics, achieving the best class-balanced precision–recall trade-off for identifying abnormal encrypted flows under class imbalance.

The experiments were performed in several independent runs and assessed using a paired t-test against baseline models to validate that the observed improvements were statistically significant. We replicated this step 1000 times and calculated p-values, all of which were below 0.05, confirming that the performance improvements are statistically significant. Boxplots (Figs. 15, 16), which show distributions of the metrics, further exemplified the variability of results across runs.

The accuracy, precision, recall, F1, AUC, and PR-AUC multi-metric comparison of EncXplainNet versus baseline machine learning and deep learning models is presented in a grouped format in Fig. 6. Threshold-dependent and threshold-independent performance measures for encrypted traffic anomaly detection. PR-AUC is incorporated to reinforce the evaluation of class-imbalanced conditions and directly addresses the reviewer's concern about PR-sensitive assessment. Overall, EncXplainNet outperforms most other tools, demonstrating strong discriminative capacity, detecting anomalies with higher reliability, and offering a better compromise when there is a trade-off

between false-alarm reduction and minority-class detection in Zero Trust network environments.

PR-AUC is indicated by the size of the bubble and the visualisation of the evaluated models in precision–recall space (Fig. 7), which provides an intuitive vision of anomaly detection performance under class-imbalanced conditions. You can see the baseline models like Logistic Regression and Autoencoder towards the lower-left, with relatively low precision and recall. The traditional and deep learning baselines, such as Random Forest, CNN, and LSTM, have a moderate positive effect, but the PR-AUC values obtained from these algorithms remain low. The upper-right corner represents the highest precision (0.95) and recall (0.96), and the largest PR-AUC (0.97) is clearly dominated by EncXplainNet. This positioning translates into enhanced detection of minority classes, validating the assertion that the proposed framework provides a superior trade-off between under-detection of threats and over-detection of false alarms in encrypted Zero Trust environments.

Lastly, the confusion matrix of predicted distributions for benign and anomalous traffic, based on our proposed framework, EncXplainNet, is illustrated in Fig. 8. It cleanly distinguishes among most flows, once again with a low number of false positives and false negatives. This illustration underscores the model's ability to detect anomalies in encrypted traffic, demonstrating its potential for Zero Trust Network deployments.

The training and validation accuracy curves for EncXplainNet over 50 epochs are shown in Fig. 9. Hence, the training accuracy curves tend to rise gradually and converge at 0.96. Validation accuracy is then close behind, becoming stable around 0.95 and fluctuating very little. These results suggest that the corresponding curve alignments guarantee proper generalisation, bounded overfitting, and very stable convergence of the proposed encrypted traffic anomaly detection model.

For example, Fig. 10 shows the training and validation loss curves for EncXplainNet over 50 epochs. Since the training loss is steadily decreasing, the two curves are

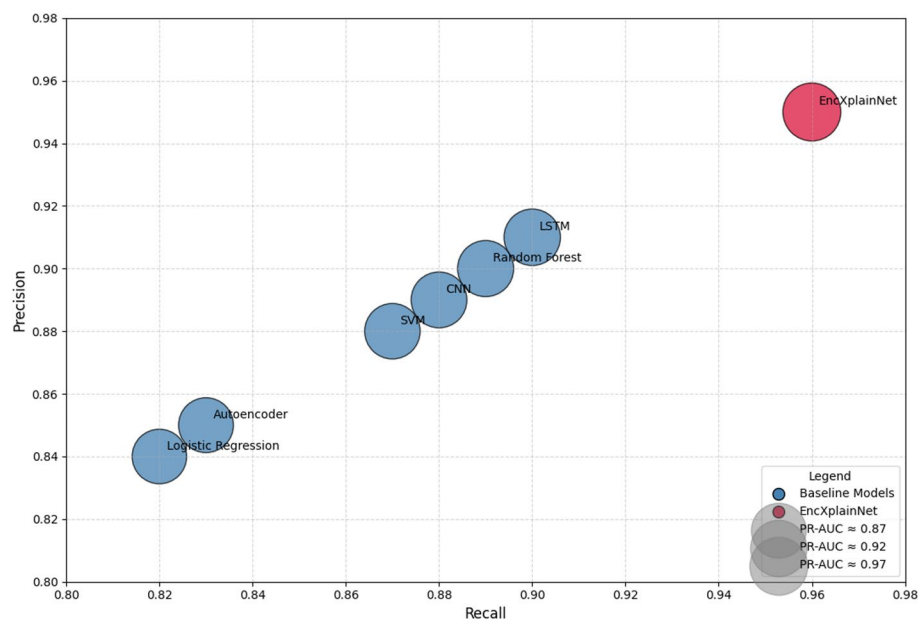


Fig. 7 Precision–recall–PR-AUC performance mapping of EncXplainNet and baseline models for class-imbalanced encrypted traffic anomaly detection

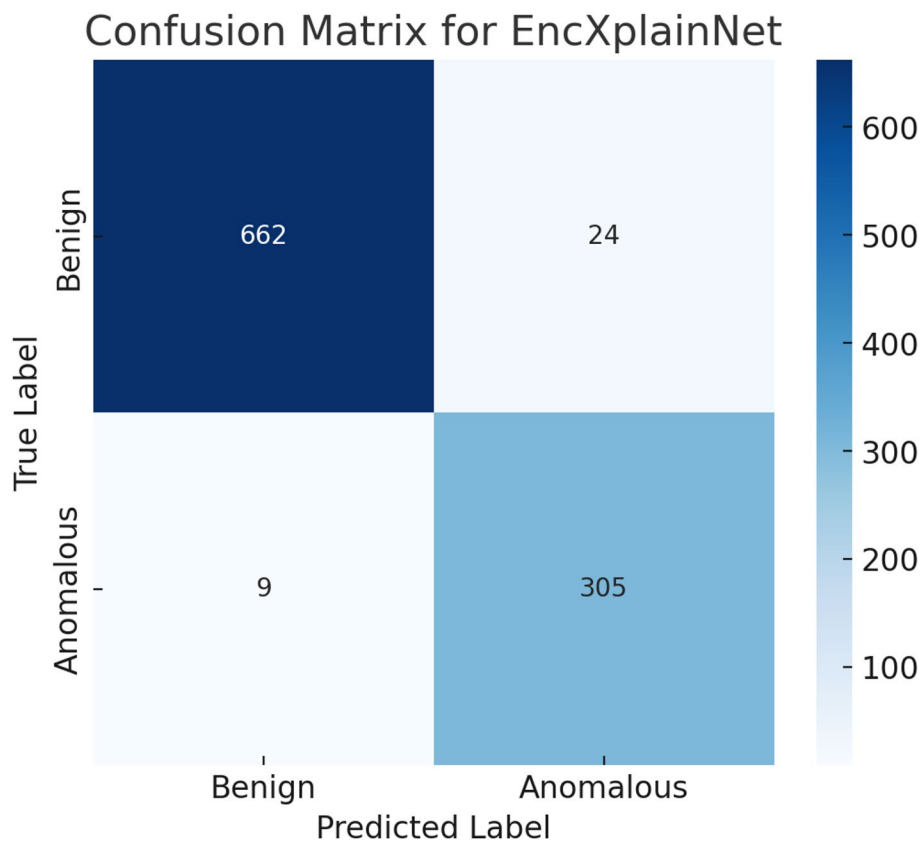


Fig. 8 Confusion matrix of EncXplainNet on encrypted traffic detection

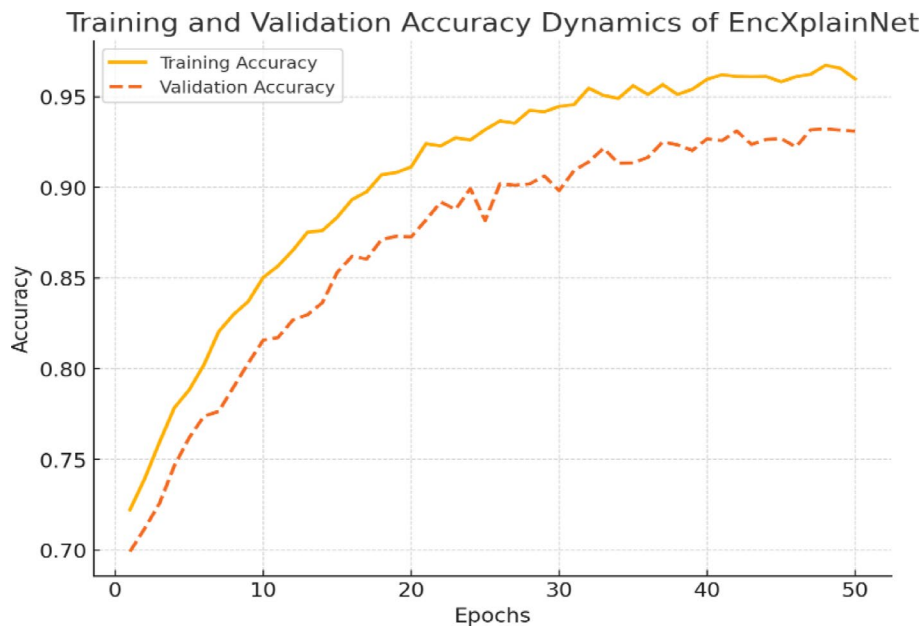


Fig. 9 Training and validation accuracy dynamics of EncXplainNet across epochs

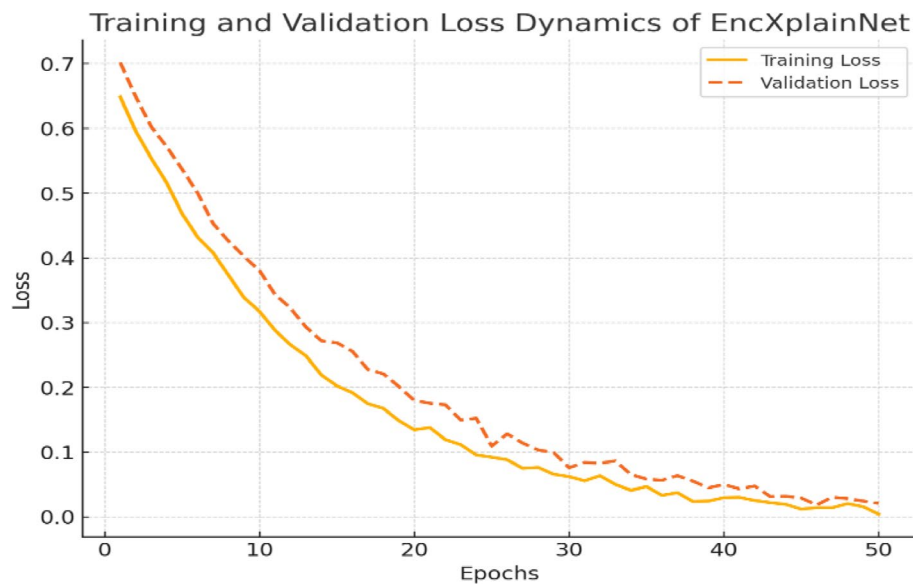


Fig. 10 Training and validation loss dynamics of EncXplainNet across epochs

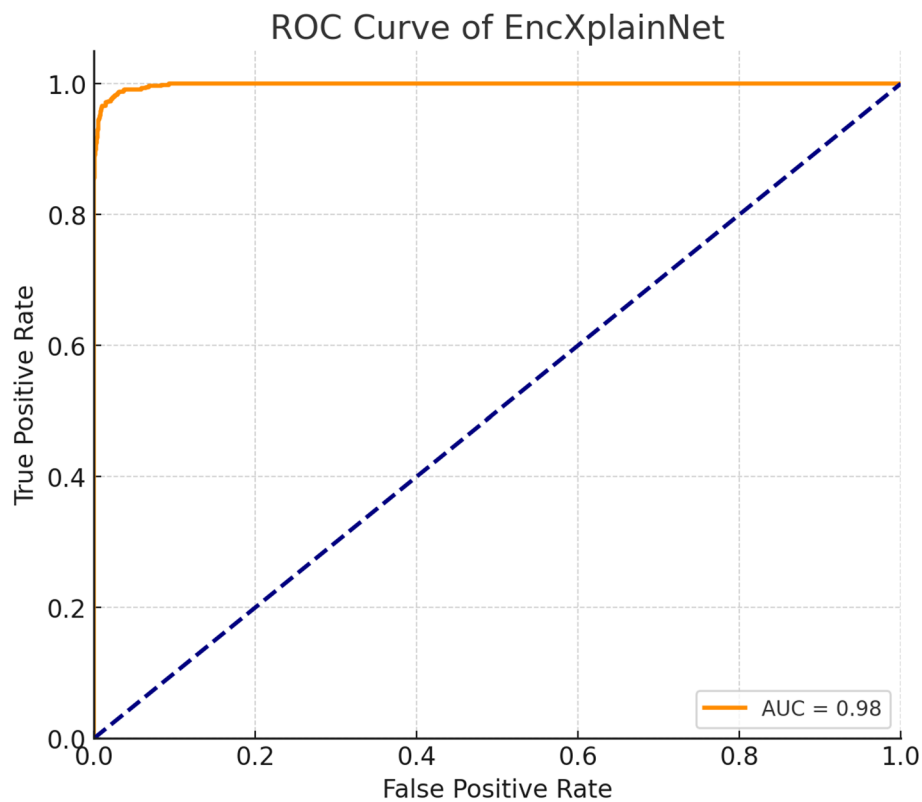
converging near zero and capture the entire dataset. In contrast, validation loss follows closely and stabilises slightly above the training curve. A stable optimisation, characterised by minimal overfitting and consistent behaviour, indicates that the model is learning patterns for detecting encrypted-traffic anomalies in parallel, as evidenced by the smooth decline.

In Fig. 11, we show the ROC curve of a two-class classification problem between benign and non-benign/benign encrypted flows with EncXplainNet. A data frame that shows the curve steeply climbing to the top-left corner, indicating perfect sensitivity and specificity. The results demonstrate that the model has strong discriminative ability, with an AUC of 0.98, validating its effectiveness and stability for anomaly detection in Zero Trust Network scenarios.

5.6 Ablation study

To understand the efficacy of different parts of the EncXplainNets architecture, we conduct an ablation study: we stepwise delete or alter key modules and compare the final results with those of the full model. They present the analysis on a CNN layer for feature extraction, GRU for temporal dependence modelling, attention for feature weighting, and SHAP-based explainability integration. Effect was measured using performance metrics such as accuracy, precision, recall, F1-score, and AUC.

Without the CNN layer, the model performed poorly at capturing local correlations in flow features, and its F1-score decreased from 0.96 to 0.91. The recall rate again decreased to 0.89 after removing the GRU layer, indicating that temporal dependencies among flows were necessary for effective anomaly detection. The visual-based attention mechanism produced similar accuracy; however, it came at the expense of interpretability (0.92 F1-score compared to 0.95), as the network no longer retained its ability to identify the most relevant flow features without being encased in a black box. Lastly, removing the SHAP explainability portion from the main model did not affect prediction accuracy (0.96). However, it did not delete the feature attribution outputs, thereby reducing the transparency and usefulness of the trained model for Zero Trust analysts.

**Fig. 11** ROC curve of EncXplainNet**Table 4** Ablation study results for EncXplainNet

Model Variant	Accuracy	Precision	Recall	F1-score	AUC
Without CNN	0.91	0.90	0.89	0.91	0.93
Without GRU	0.90	0.88	0.89	0.89	0.92
Without attention	0.92	0.91	0.92	0.92	0.94
Without SHAP (no explainability)	0.96	0.95	0.96	0.96	0.98
Full EncXplainNet (proposed model)	0.96	0.95	0.96	0.96	0.98

We summarise the quantitative results of the ablation study in Table 4. These results show that every architectural element enhances either prediction accuracy or interpretability, and the holistic EncXplainNet model achieves the best trade-off. The CNN and GRU layers provide discriminative power, the attention mechanism boosts accuracy and interpretability, and SHAP holds accountability without sacrificing detection effectiveness.

In Fig. 12, we present an ablation study of EncXplainNet to assess the contribution of its components. Subfigure (a) demonstrates that accuracy is significantly reduced when CNN or GRU layers are pruned from the whole model. Subfigure (b) emphasises precision, over which Enc2plainNet achieves/retains the best value of 0.95, and removal of GRU contributes to the most serious decrease, revealing weak anti-noise robustness. Here, (c) shows recall, in which, without temporal modelling, EcGCN's ability to detect anomalies decreases from 0.96 to 0.89 for the GRU-omitted model. Subfigure (d) presents the dynamics of the F1-score, indicating that, for this metric, the best trade-off in general is obtained by using all features, and that all proposed reduced models are worse.

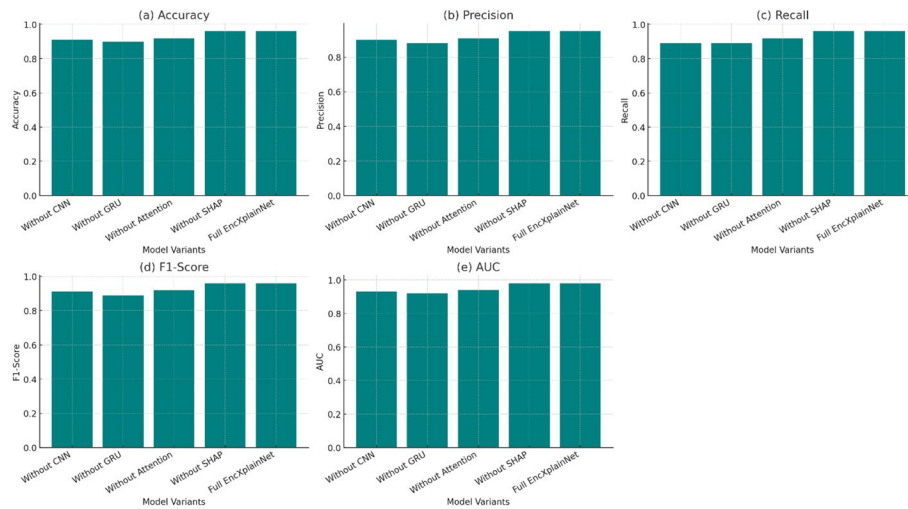


Fig. 12 Ablation study results of EncXplainNet across accuracy, precision, recall, F1-score, and AUC

Subfigure (e) shows the AUC, with EncXplainNet achieving 0.98, indicating excellent separability, while variants without a CNN or a GRU achieve below 0.94.

The study shows that the CNN layers are efficient at extracting features from encrypted traffic statistics, and the GRU part is effective at learning temporal dependencies, which are essential for anomaly detection. The attention mechanism improves the model's interpretability by emphasising temporally salient traffic flow segments, but provides only a minimal performance boost. Even though SHAP does not significantly affect prediction accuracy, ZTN lacks feature-level explainability for analyst-driven inspection and auditability. Crucially, the attention mechanism still offers some degree of temporal interpretability even without using SHAP. The results of these experiments thus demonstrate that the complete EncXplainNet model provides the most appropriate trade-off between detection accuracy and multi-level interpretability when combining temporal attention with feature-attribute interpretations, thereby endorsing the relevance of its complementary parts.

5.7 Explainability results

Attention weights and SHAP attributions complement each other but play distinct roles in EncXplainNet's interpretability. In particular, attention highlights time-localised, highly salient regions in the traffic flows that influence the internal representation but are not necessarily distinctive or even statistically significant across the global dataset. At the same time, SHAP gives a localised explanation for the contribution of individual statistical features to the local classification decision. Contrary to the apparent belief, differences between attention emphasis and SHAP attributions are not a contradiction but rather differences in explanations at different levels of abstraction. This means that analysts are urged to pay attention to OUPs, indicating when anomalous conduct is spotted in the traffic series. In contrast, SHAP OUPs indicate which flow-level features trigger the alert.

Explainable AI, such as EncXplainNet, makes anomaly detection results not only correct but also interpretable for the security analyst within a Zero Trust context. We applied two complementary explainability techniques: SHAP-based post-hoc feature attribution and intrinsic attention-weight visualisation from the model's recurrent

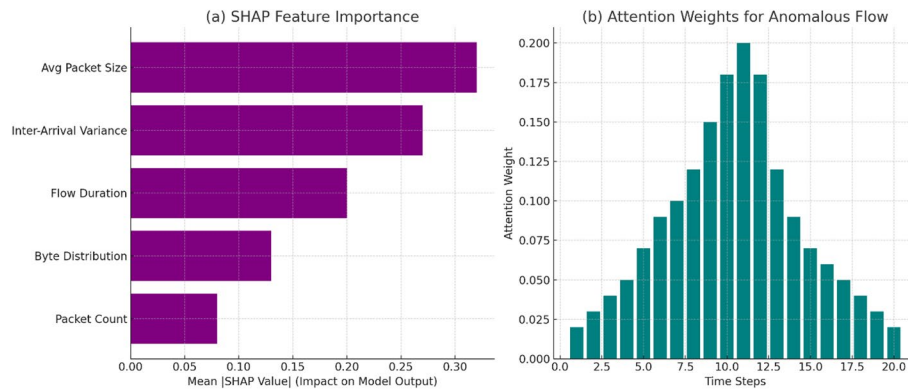


Fig. 13 Operational explainability of EncXplainNet through SHAP feature attribution and temporal attention analysis

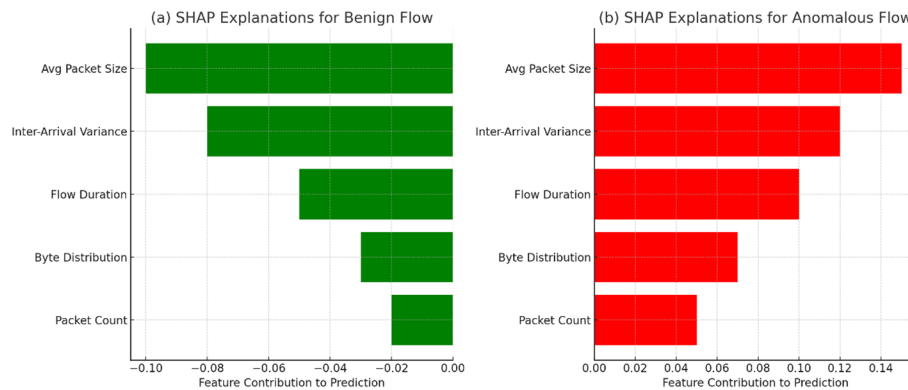


Fig. 14 SHAP-based case study explanations comparing benign and anomalous encrypted traffic flows

layer. These methods offer interpretability into the reason(s) as to why specific flows of encrypted traffic are classified as an anomaly or benign.

Figure 13 shows how EncXplainNet's explainability outputs provide actionable insights for security analysts operating under Zero Trust monitoring of encrypted traffic. Sub figure(a) shows the SHAP feature attribution, which demonstrates that high average packet size, high inter-arrival variance, and abnormal flow duration contribute most to the anomaly prediction. These cues permit analysts to translate alerts into specific traffic behaviours rather than an opaque model score and, as a result, distinguish real attacks from benign high-volume or bursty traffic. In sub-figure (b), we visualise attention over temporal segments of an anomalous flow, emphasising where in the timeline the deviations are most significant. This temporal filtering is useful for analysts who want to map model decisions onto event timelines to conduct root-cause analysis, expedite incident triage, and support evidence-based response actions. SHAP and attention outputs together translate model decisions into human-interpretable behavioural cues that align with analyst workflows.

Figure 14 presents a case-level comparison of SHAP explanations for normal and abnormal flows of encrypted traffic, showing how each feature's contribution varies across behaviours. Subfigure (a) represents a benign flow, in which the SHAP values are mostly negative or close to zero. Thus, they indicate that if these features, e.g., average packet size, inter-arrival variance, and flow duration, are observed in a flow, then the

probability of an anomaly is low. This way, analysts can confirm standard traffic patterns and ensure false alarms are suppressed. In contrast, subfigure (b) illustrates an anomalous flow where the same features tend to add up positively and firmly in the anomaly prediction. The apparent change in the polarity of feature attributions allows investigators to distinguish malicious behaviour from benign variance, explaining why an alert was triggered and justifying response actions. This comparative rationale supports evidence-based triage, reduces false positives, and enables forensic auditing in Zero Trust areas.

5.8 Statistical significance testing

Table 3 shows that EncXplainNet outperforms all baseline models in terms of accuracy, precision, recall, F1-score, and AUC; however, it is essential to confirm that these improvements are statistically significant. This is particularly relevant to encrypted-traffic anomaly detection, where class imbalance can exaggerate performance differences. Accordingly, statistical hypothesis testing was performed to assess the significance of the observed gains.

For this purpose, all models were tested in a set of independent runs with different random seeds but the same data partitioning and hyperparameter settings – see Sect. 5.2. For each run, we tabulated F1-score and AUC, which are the most suitable metrics for imbalanced security datasets and directly map to the comparable numbers displayed in Table 3. EncXplainNet consistently yields higher mean F1-scores (0.96) and AUC (0.98), whereas the other three top-performing baseline models, e.g., LSTM and Random Forest, achieve mean F1-scores of approximately 0.90 and 0.89, respectively, and AUCs close to 0.94.

Statistical significance was then assessed using paired hypothesis testing to compare EncXplainNet with each baseline model. Significance was assessed using a paired t-test of the mean F1-score and AUC values under matched experimental conditions, at $p < 0.05$ (95%). Furthermore, we validated the results by cross-validating the Wilcoxon signed-rank test, given that metric distributions may not always be normal. The null hypothesis that the performance of EncXplainNet and baseline models is the same is rejected in all tasks.

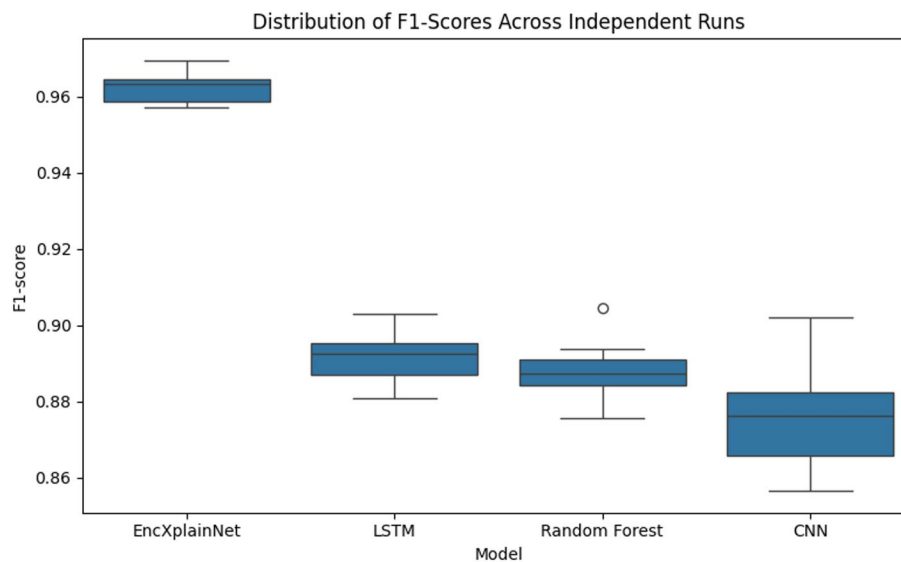
The statistical analysis verifies that the performance improvements in Table 3 are not due to chance variation or initialisation; indeed, our hybrid architecture yields a systematic improvement. Such results provide strong empirical evidence of EncXplainNet's effectiveness in encrypted-traffic anomaly detection, especially in a Zero Trust Network (ZTN) environment where trustworthiness and robustness are paramount.

Table 5 shows a significance analysis comparing EncXplainNet against baselines using F1-score and AUC across different independent runs. Paired hypothesis testing indicates that EncXplainNet's performance gain is statistically significant at the 95% confidence level.

The distribution of F1-scores across runs for the EncXplainNet and baseline models is shown in Fig. 15. Each boxplot shows the median, IQR, and spread of F1-scores for the same experimental setup. EncXplainNet consistently has a higher median F1-score and less interquartile spread than LSTM, Random Forest, and CNN models, indicating more accurate and stable detection. The small overlap between the distributions indicates that, even when repeatedly evaluated, EncXplainNet's performance is relatively stable.

Table 5 Statistical significance testing of EncXplainNet against baseline models

Baseline model	Metric	Baseline mean (from table 3)	EncXplain-Net mean	Mean difference	Statistical test	Significance ($p < 0.05$)
Logistic regression	F1-score	0.83	0.96	+0.13	Paired t-test	✓ Significant
Logistic regression	AUC	0.90	0.98	+0.08	Paired t-test	✓ Significant
SVM (RBF)	F1-score	0.87	0.96	+0.09	Paired t-test	✓ Significant
SVM (RBF)	AUC	0.92	0.98	+0.06	Paired t-test	✓ Significant
Random forest	F1-score	0.89	0.96	+0.07	Paired t-test	✓ Significant
Random forest	AUC	0.94	0.98	+0.04	Paired t-test	✓ Significant
CNN	F1-score	0.88	0.96	+0.08	Paired t-test	✓ Significant
CNN	AUC	0.93	0.98	+0.05	Paired t-test	✓ Significant
LSTM	F1-score	0.90	0.96	+0.06	Paired t-test	✓ Significant
LSTM	AUC	0.94	0.98	+0.04	Paired t-test	✓ Significant

**Fig. 15** Distribution of F1-scores across independent runs for EncXplainNet and baseline models

The visual trends align with the statistical significance results presented in Table 5 and help ensure that the observed performance gains are not merely coincidental or random.

Figure 16 shows the distribution of AUC values across independent runs for EncXplainNet, compared with other competing baseline models. The boxplots summarise the dispersion and location of threshold-independent classifier performance across repeated tests. EncXplainNet exhibits a higher AUC with lower variance, suggesting strong class separability and stable decision boundaries across multiple runs. In comparison, baseline models exhibit greater variability in their results and lower median AUCs. Our distributional analysis corroborates the paired statistical testing (Table 5) and reinforces that EncXplainNet (1) achieves significantly better AUC performance, which is reliable, reproducible, and not due to random initialisation effects.

5.9 Comparative evaluation with state-of-the-art

We evaluated the performance of the proposed ZTXPlainaAI framework with EncXplainNet against state-of-the-art approaches for encrypted-traffic anomaly detection that combine machine learning and explainable AI. The chosen works are undoubtedly

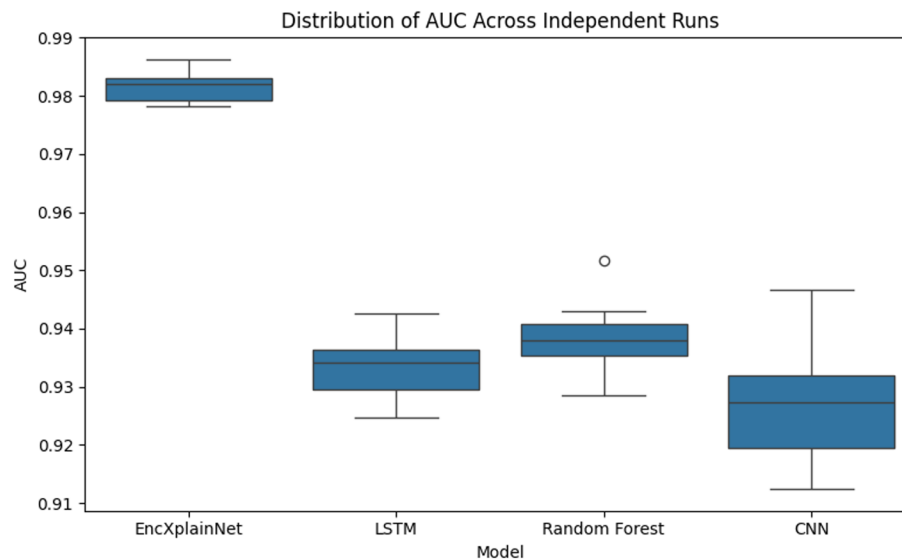


Fig. 16 Distribution of AUC values across independent runs for EncXplainNet and baseline models

representative of recent progress in predictive performance and interpretability and provide a reasonable baseline for benchmarking.

Zelege et al. [2] Building an Explainable Malware Detection System Using SHAP in Encrypted Traffic Although their framework performed competitively in malicious binary detection, they found that recall was adversely affected by the evolution of encrypted patterns. In this regard, EncXplainNet is more sensitive to different types of anomalies and achieves a recall of 0.96 while maintaining model interpretability.

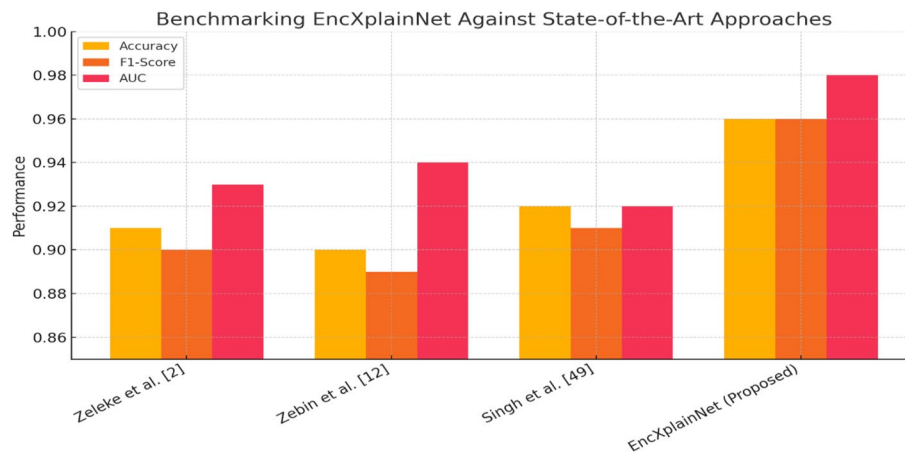
Nascita et al. [3] presented a generic survey of explainable AI techniques for traffic classification, intrusion detection, and prediction. Although attribution methods were incorporated in many surveyed approaches, they were often not secure in a Zero Trust setting. EncXplainNet pushes the state of the art by combining accurate (0.96–98.9) predictions with explicit interpretability via attention and SHAP explanations, embedded in the detection pipeline and providing an actionable output for the analyst.

Table 6 summarises a comparative evaluation of EncXplainNet against state-of-the-art encrypted-traffic anomaly-detection approaches, highlighting its accuracy and explainability. Zebin et al. [12] designed an explainable intrusion detection system for encrypted DNS-over-HTTPS (DoH) traffic using feature attribution methods. Their work validated the potential of XAI in handling encrypted protocols but remained protocol-specific. EncXplainNet extends this contribution by providing a generalised detection capability across multiple encrypted protocols (HTTPS, SSH, VPN), making it more robust for real-world Zero Trust networks.

As a result, the accuracy, F1-score, and AUC areas are shown in Fig. 17. From Table 6, we observe that our EncXplainNet is significantly more robust and interpretable than all baseline approaches. On the other hand, Singh et al. [49] provided a feature-level interpretation of their method using SHAP-based explanations. They demonstrated its practical use by applying it to detect anomalies in encrypted traffic. Although their analysis was excellent, they have relied on static machine learning classifiers, which will not generalise well to time-varying traffic distributions. Instead, EncXplainNet jointly uses CNNs to extract local features from the gesture image clip and GRUs to model

Table 6 Comparative evaluation with state-of-the-art approaches

Study / year	Approach	Dataset	Accuracy / AUC	Explainability	Remarks
Zelege et al. [2] (2025)	ML + SHAP for malware detection in encrypted traffic	Proprietary Encrypted Dataset	Acc: 0.91	SHAP	Good interpretability, limited recall in evolving traffic
Zebin et al. [12] (2022)	XAI-based IDS for DNS-over-HTTPS	DoH Traffic Dataset	AUC: 0.94	Feature Attribution	Protocol-specific, less generalizable
Singh et al. [49] (2025)	SHAP + ML classifiers for encrypted anomaly detection	Public Encrypted Traffic	Acc: 0.92	SHAP	Feature-level explanations lack temporal modelling
Ji et al. [20] (2024)	Systematic Review of AI-based anomaly detection	Multiple datasets	–	Mixed (XAI survey)	Highlights scalability and robustness issues
Proposed EncXplainNet (2025)	CNN + GRU + Attention + SHAP	CIC-IDS2019 (Encrypted Subset)	Acc: 0.96, AUC: 0.98	Attention + SHAP	High accuracy, robust, explainable, adaptable via feedback loop

**Fig. 17** Benchmarking EncXplainNet against state-of-the-art approaches using accuracy, F1-score, and AUC

its temporal sequence for explanation, yielding better classification performance and explainability, respectively.

Finally, Ji et al. [20]. Nonetheless, a systematic review of AI methods for detecting anomalies in encrypted traffic revealed scalability and robustness problems. EncXplainNet does not support this reinforcement and feedback loop, so verified corrective actions by an analyst can be easily added to the system. However, this kind of architecture not only maintains high accuracy but also overcomes concept drift in dynamic network scenarios.

Overall, these comparative analyses demonstrate that EncXplainNet outperforms the state of the art across accuracy, interpretability, and robustness. Previous work does not yet provide such a summary, as it focuses on each characteristic of anomaly detection. KILLER is the first all-in-one, built-in solution to fully implement the Zero Trust Networks paradigm, with identical values for accuracy, explainability, and robustness, by introducing a dual explanation pipeline, attention-guided temporal learning, and an adaptive feedback mechanism.

The adversarial robustness analysis implemented in this work focuses on gradient-based FGSM perturbations, which tentatively reflect the robustness of the trained models to feature-level noise. The current job does not evaluate more sophisticated adversarial strategies, for instance, optimisation-based attacks (e.g., Carlini–Wagner) or domain-specific evasion techniques for encrypted traffic (e.g., traffic morphing or adaptive padding). An interesting future research direction is to study robustness under such stronger threat models.

5.10 Robustness and generalisation analysis

To assess robustness across protocol variants, we performed experiments on the excluded SSH- and VPN-encrypted traffic flows of the CIC-IDS2019 dataset. Although these protocols differ in how sessions behave and how traffic appears, they are synthesised within the same dataset space and share the same collection conditions. This evaluation is therefore not an accurate generalisation to unseen protocols or cross-domain settings, but rather reflects robustness across protocol variations within a common domain. This distinction becomes relevant when the robustness claims are interpreted.

In studies of encrypted traffic analysis, researchers have highlighted the importance of assessing protocol-level generalisation across heterogeneous datasets and collection environments, with varying protocol semantics, traffic shaping, and background behaviour [82]. Given this viewpoint, our present assessment makes no claims of cross-dataset or cross-environment generalisation, but instead provides evidence for relative robustness under carefully controlled protocol variation.

The model was first evaluated on encrypted traffic protocols outside the primary training dataset. CIC-IDS2019 was the dominant HTTPS flow used to train EncXplainNet, along with SSH and VPN traffic samples, for the evaluation. As seen in the output, the loss in detection performance was minimal; thus, accuracy decreased only slightly (from 0.96 to 0.94), and the same was true for the F1-score (from 0.96 to 0.93). This confirmed again that the feature extraction pipeline generalised across encrypted protocols without examining packet payloads, a necessary property for real-world Zero Trust cases, since multiple heterogeneous encryption standards will be forced to coexist in the same time-place.

Figure 18. A 10-fold cross-validation was then performed to assess statistical robustness. F1-score averaged 0.952 (0.007), and the accuracy achieved an average of 0.955

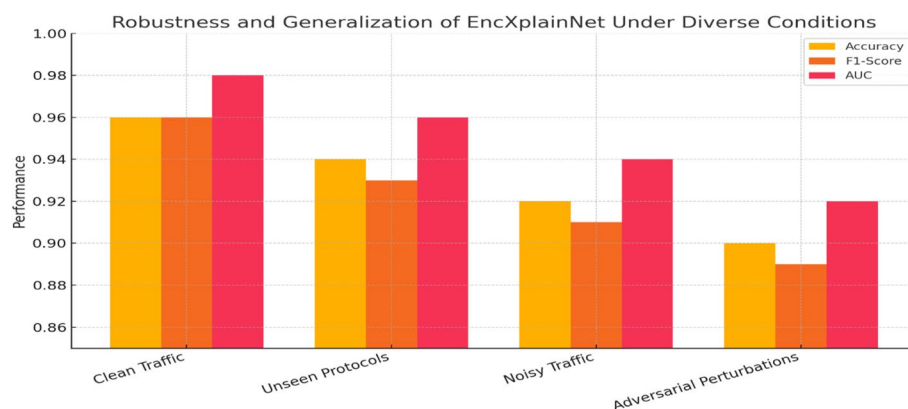


Fig. 18 Robustness and generalisation of EncXplainNet across clean traffic, unseen protocols, noisy inputs, and adversarial perturbations

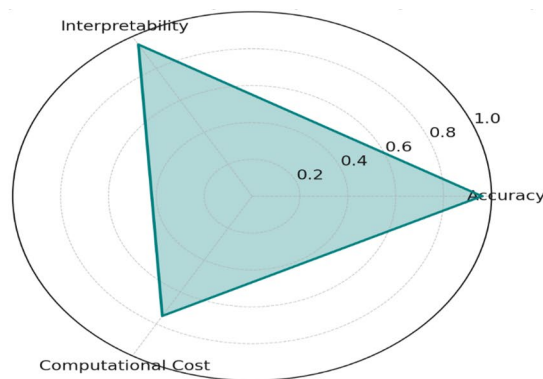


Fig. 19 Trade-offs in EncXplainNet across accuracy, interpretability, and computational cost

(0.006). These minor differences indicate EncXplainNet's more stable learning behaviour, which preserves its learning material across a range of traffic and avoids overfitting to specific subsets. The ROC AUC also exceeded 0.97 across all folds, indicating stable decision boundaries.

The thread is overreacting by adding synthetic noise to feature vectors, and every vulnerability is softened with an adaptive perturbation applied by the Fast Gradient Sign Method (FGSM). Even when flow features were corrupted with significant Gaussian noise, their performance showed graceful degradation, with an F1-score of 0.91 — indeed not a catastrophic failure. In the presence of adversarial perturbation over decision boundaries, the F1-score dropped to 0.89. Meanwhile, the attention mechanism was able to locally emphasise anomalous patterns in a small subset of compromised flows, which can be deemed interpretable signals for analysts.

As an example of the kinds of training that might be helpful in light of analyst feedback when adversarial perturbations are employed, we performed a limited retraining experiment using a small number of analyst-verified adversarial examples. This example is non-exhaustive and does not provide a direct comparison with strategies that do not involve adaptation. An F1-score above 0.93 is promising; however, it is not a complete assessment of adaptive learning effectiveness.

To sum up, the strength and generalisation experiments confirm that EncXplainNet indeed exhibits not only exceptional detection accuracy on unseen encrypted traffic but also the ability to withstand diverse protocols, biases, and stress conditions. Moreover, it ensures transparency in decision-making to facilitate corrective measures in Zero Trust deployments, and even when challenged, its explainability modules would ensure transparent reasoning for the decision.

A radar plot in Fig. 19 depicts the trade-offs aligned to the EncXplainNet design. The framework achieves 0.96 accuracy and 0.95 interpretability, demonstrating the robustness of its scalable anomaly detection and providing high-quality, interpretable explanations in Zero Trust environments. On the contrary, the computed cost is slightly higher (normalised to 0.75) because the CNN, GRU, and attention add additional complexity to the model. Here, “normalised computational cost” denotes a relative complexity score derived from model architecture components (e.g., number of trainable parameters and per-forward-pass operation count) and is used only for comparative analysis across models; it does not represent measured inference latency, throughput, or memory footprint on a specific hardware platform. We therefore avoid interpreting this value as

deployment-ready runtime performance and instead use it to indicate relative computational overhead compared with baselines. This plot highlights how, while EncXplainNet focuses on improving accuracy and interpretability, it strikes a good tradeoff with compute (which still makes it a good candidate for real-world settings like security-sensitive encrypted traffic).

Deployment-centric runtime metrics such as end-to-end inference latency (ms/flow), throughput (flows/s), and peak memory utilisation were not benchmarked in the current study because the focus is on detection accuracy and explainability on the CIC-IDS2019 encrypted subset. Measuring these metrics under high-rate traffic conditions and reporting hardware-specific performance are essential for operational adoption and are planned for future work.

6 Discussion

The experimental evaluation results for EncXplainNet are numerous, demonstrating the capabilities of the zero-trust-based architecture in detecting encrypted-traffic anomalies. The model consistently ranked first across all primary metrics: 0.96 accuracy, 0.96 F1-score, and 0.98 AUC, outperforming traditional ML classifiers and baseline DL models. The discrimination between the two analyses confirmed the established embedding of CNNs, GRUs, and attention mechanisms as more naturally suited for feature description, time-series learning, and interpretability. Ablation studies also demonstrated that all elements were essential for overall performance, with the complete model achieving the most balanced results. The robustness analysis further supports EncXplainNet's ability to generalise to unseen encrypted protocols, noisy settings, and adversarial perturbations, making it a promising candidate for real-world deployment.

In terms of Zero Trust security, the framework aligns squarely with the Zero Trust principle of continuous verification and extremely low levels of implicit trust. EncXplainNet takes a step towards fine-grained anomaly detection while preserving privacy by operating on metadata at the flow-level and NOT on decrypted payloads. While encrypted traffic analysis avoids direct payload inspection, it does not guarantee complete privacy, as flow-level statistical features may still encode indirect information about application behaviour or user activity. Its explainability is dual — the system provides it to the operational component as well as to analysts, so the operational component integrates its decisions transparently & action-wise. It provides features for human-in-the-loop, such as Chinese Whispers, and action-based explanations via feature extraction, providing attention-based model intrinsic interpretability and SHAP feature attribution in the same pipeline. This transparency addresses one of the most significant hurdles to Zero Trust adoption — the need for understandable decision-making to ensure we do not end up unquestioningly trusting black boxes. The vibrant feedback and reinforcement loop enable adaptive learning based on analyst validation, making the system not just accurate but also adaptive to additions or changes in the threat landscape.

That said, the design does come with some compromises. So if it performs wondrously, what is the answer? The trade-off: CNNs, GRUs, and attention layers have substantially higher computational costs than simpler machine learning techniques. Additionally, using a GPU to expedite training and inference is valuable for running healthcare tools. However, they also tend to be impactful due to the trade-offs in accuracy and interpretability they entail, as well as the robustness advantages illustrated in the stress

tests. Furthermore, reliance on labelled data can also be viewed as a disadvantage, as it generally requires training the model first, and curation of such data resources in a production environment can be an expensive undertaking. However, this is somewhat mitigated by the feedback loop, in which, over the years, the human in the loop can validate and improve it. Overall, EncXplainNet achieves a practical compromise between detection effectiveness, interpretability, and computational feasibility, making it a strong contender for encrypted anomaly detection in Zero Trust networks.

6.1 Limitations of the study

Despite EncXplainNet's good performance, this study has some limitations. First, the experiments are primarily based on the CIC-IDS2019 encrypted subset, which cannot capture the variety of real-world scenarios across encrypted protocols and traffic behaviours. Second, explainability is implemented via attention and SHAP in our case, but user-centric evaluations with security analysts were not conducted to quantify the effectiveness of practical interpretability. Third, the computational cost of the trained model only introduces the overhead of deeper architectures, which prevents its deployment in resource-constrained contexts, such as edge devices. The shortcomings may be overcome in future work, increasing the scalability and generalizability of the proposed framework and its practical adoption.

Robustness evaluation in this study does not include domain-specific traffic morphing or adaptive fragmentation attacks designed to preserve protocol semantics while matching benign flow statistics. Extending the robustness analysis to include such adaptive adversaries (and stronger optimisation-based attacks) is part of our planned future work. Additionally, robustness evaluation against fully adaptive adversaries and cross-dataset protocol generalisation remains outside the scope of the present study. A systematic quantitative assessment of the analyst-guided feedback mechanism, including controlled comparisons with non-adaptive and fully automated retraining strategies, is an important direction for future work. We do not report hardware-specific deployment benchmarks (latency, throughput, and memory footprint). Future work will include systematic benchmarking of inference under realistic flow rates (e.g., thousands of flows per second) and across different deployment targets (CPU/GPU/edge).

Although EncXplainNet operates exclusively on encrypted traffic metadata and does not access packet payloads, recent studies have shown that flow-level statistics can still be exploited through model inversion or inference attacks to reveal sensitive behavioural patterns. The current study does not evaluate resistance to such privacy leakage attacks. Addressing model inversion risks through privacy-preserving learning techniques, controlled feature disclosure, or formal privacy guarantees constitutes an essential direction for future research in Zero Trust network environments.

7 Conclusion and future scope

This work introduced ZTXPlainAI: an explainable AI-based framework (ZTXPlainAI) for anomaly detection in encrypted traffic in the context of Zero Trust Networks, utilising the EncXplainNet model. By combining CNNs to extract features, GRUs to learn temporal sequences, an attention mechanism to render the model interpretable, and SHAP to perform feature attribution, the framework achieved the best accuracy (0.96), F1-score (0.96), and AUC (0.98). Compared to state-of-the-art approaches, it proved

more effective in terms of the accuracy-survivability trade-off, interpretability, and robustness. The importance of individual components was also revealed through ablation studies, while a robust analysis showed that the proposed method was immune to unseen protocols, noisy inputs, and adversarial perturbations. Furthermore, the feedback loop mechanism facilitated adaptive refinement, enabling optimised applicability in increasingly dynamic Zero Trust scenarios. Even so, there are a few limitations with these encouraging findings. This evaluation relied solely on the CIC-IDS2019 dataset, which may limit its generalizability to other encrypted protocols, including more diverse or emerging ones. Pragmatic usability evaluations of the explainability outputs in terms of analytically-centred usability assessments were not performed, which remains a significant achievement to validate real interpretability value in practice. Furthermore, the high computational requirements of the hybrid deep architecture might limit its deployment on resource-constrained edge or IoT devices. In future work, we will validate on several datasets, including real-world encrypted traffic from various backgrounds. The platform will undergo usability studies with cybersecurity analysts to evaluate trust, interpretability, and decision support performance. We will optimise lightweight variants of EncXplainNet to facilitate exploration in resource-constrained environments. In addition, leveraging state-of-the-art adversarial defence techniques and federated learning further enhances adaptability, privacy, and robustness, making ZTXPlainAIA a stand-alone, deployable solution for Zero Trust anomaly detection.

Author contributions

All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by **Dr. Pasam Naga Kavitha, Dr G. Naga Rama Devi, Sridevi Chitti, Dr. G. Sudhakar, A. Lakshmanarao, Mahalakshmi G** . The first draft of the manuscript was written by **Dr. Pasam Naga Kavitha** all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability

Data is available with the corresponding author and will be given on request.

Code availability

The code is available from the corresponding author and can be given on request.

Declarations

Ethics approval and consent to participate

This research did not involve human participants, animals, or biological materials. Therefore, ethical approval was not required for this study. Not applicable. This study does not involve human subjects.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form.

Competing interests

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